

SHARD: cell-keyed residual splitting for alignment-resistant private dense retrieval

Sergey Kurilenko

Moscow Institute of Physics and Technology
sergkurilenko@gmail.com

July 12, 2026

Abstract

Dense retrieval systems expose document geometry when vector stores are compromised, and a global protective transform can often be aligned from known pairs. We study SHARD, which splits PCA coordinates into a short routing prefix and a residual protected by independent cell-local orthogonal keys. It supports CKKS ciphertext–plaintext reranking but is evaluated as a leakage trade-off, not a cryptographic document-privacy guarantee.

Corrected scoring uses $V^\top(x - \mu)$ for documents and $V^\top q$ for queries, preserving raw ranking up to a query-dependent constant. Across ten BEIR/MIRACL configurations it reproduces raw nDCG@10 and recall, whereas centring both sides loses up to 0.080 nDCG. Cell keys spread diffuse known-pair evidence across compartments, but minimum-norm alignment recovers useful signal far below full key rank, so there is no hard de-anonymisation threshold.

Real CKKS confirms a maximum score error of 2.29×10^{-6} and no top-1 flips; block packing cuts query upload by 74–87% but raises in-process p50 latency by 14–26%. In a strengthened GTR case, an unknown key lowers token-F1 from .665 to .242; a wide prefix and eight pairs restore much. Under 25–90% release overlap, the unchanged prefix and clean residual norm link persistent rows with R@1 at least 0.9996, although cell-Gram linkage degrades under heavy churn. A formally calibrated Gaussian release gives nDCG@10 at most 0.011 at $\varepsilon = 1$; its only three strict utility matches occur at $\varepsilon = 32768$ with linkage R@1 at least 0.995. SHARD therefore preserves retrieval and compartmentalises alignment evidence, but does not provide DP, unlinkability, or cancellable templates.

Keywords: dense retrieval, embedding privacy, alignment attacks, cross-release linkage, homomorphic encryption, CKKS, block-SIMD.

1 Introduction

The privacy of vector representations of text—“embeddings”—has become a first-class concern in production retrieval systems. Dense retrievers (DPR [5], ColBERT [6]) and state-of-the-art encoders (Sentence-BERT [1], multilingual-E5 [2], GTR [3], BGE-M3 [4]) push the per-document representation through a $\sim 10^2$ -dimensional bottleneck that nevertheless preserves enough information to reconstruct the original text with high BLEU [48]. VEC2TEXT [7] reports BLEU $\approx 97.3\%$ on 32-token inputs when the attacker has query access to the encoder; GEIA [8] and TEIA [9] show that even snapshot attacks (an adversary with read access to the vector database) recover personally identifiable information at high rates. For Retrieval-Augmented Generation pipelines whose vector stores typically contain customer support tickets, internal corporate documents and personal data, this rule of thumb—“a leak of the embedding is a leak of the text”—is now well documented [49, 18].

The defensive landscape is bracketed by two extremes that, individually, do not satisfy industrial requirements. Differential privacy [22, 24] adds calibrated noise per-record but, in

the naive coordinate-Gaussian formulation on retrieval-quality embeddings, it faces a difficult operating-point choice: mild noise preserves much of the nearest-neighbour geometry, while stronger noise destroys ranking quality long before the privacy budget reaches a level considered useful. Fully homomorphic encryption [28, 35] allows arbitrary computation on ciphertexts, but the ciphertext-ciphertext (ct-ct) regime is too expensive for top- k search across 10^6 documents.

The weak axis of global-linear defences. Between the two extremes sits a popular lightweight middle ground: compress the collection onto a data-dependent SVD subspace, apply a single secret orthogonal rotation, publish an approximate-nearest-neighbour (ANN) index, and rerank the encrypted query under CKKS. We first measure this global-linear stack and confirm that its protection lives on *one weak axis*—the protected store is a single globally-aligned geometry. Concretely (Section 7): a known-plaintext orthogonal Procrustes attack recovers the secret rotation with about $k = d/2$ anchors, after which an overlapping reference corpus yields near-exact paragraph recovery (99.8% top-1); the public product-quantisation index preserves a mean cosine of 0.95 and 67% of exact top-10 neighbours; and a truncated rerank can lose quality on real IR. A corrected scoring audit in this paper separates two effects that the earlier implementation conflated: centring the query as well as the documents costs 0.9–7.4 nDCG points, while the remaining half-PCA truncation gap ranges from approximately 0 to 5.3 points across the evaluated BEIR and MIRACL cells. These are exactly the weaknesses that modern few-shot alignment [10], zero-shot inversion [11] and unsupervised cross-space translation [17] attacks exploit: a single global map is, in the end, alignable.

SHARD: an attack-aware geometric transform. We introduce SHARD, a family of retrieval-preserving protective transforms designed against the alignment and index-leakage attack surface rather than against a distortion surrogate such as σ_{rec} (which, as we and others show, is not a privacy metric). The construction (Section 6) keeps the CKKS/ct-pt machinery of the baseline but replaces its single global geometry. The construction itself is quick to state. We rotate the centred embedding and split it into a short *public* prefix u —the top- d_{pub} PCA directions, which carry coarse stage-1 retrieval—and a *private* residual r . The residual is sharded into C coarse cells defined by u , and each cell is turned by its own secret orthogonal key H_c . The experiments sample dense Haar-like keys by Gaussian QR, so that the server only stores $z_i = H_{c(i)}r_i$. Reranking then happens under CKKS in the residual space, where the per-cell keys cancel, $\langle H_c r_q, z_i \rangle = \langle r_q, r_i \rangle$ in plaintext algebra; CKKS then evaluates the same identity approximately. The single knob C interpolates between a full-dimensional single-residual-key control ($C=1$) and per-document micro-keys ($C=N$).

The geometric evaluation begins with two constructive findings. First, the corrected full-dimensional split score preserves raw ranking on all ten BEIR/MIRACL encoder–dataset cells; in the plaintext experiment, the remaining measured utility gap comes from stage-1 shortlist recall. Second, cell keys compartmentalise known-pair evidence. A diffuse leak gives each cell fewer anchors, so the recoverable anchor-span rank and the resulting cosine/re-identification curve grow more slowly as C increases. This effect is continuous, however: minimum-norm linear alignment extracts useful signal far below full rank, so there is no d_{priv} privacy barrier and no claim that subthreshold records remain unidentified.

The limiting findings are just as important. The short prefix exposes less global neighbour structure than the baseline’s public half-space, but the server still sees a meaningful coarse index and exact within-cell Gram geometry. Independent re-keying therefore does not make snapshots unlinkable. At $N = 10,000$, a row-wise cross-key cosine test is near chance, yet norm assignment, Gram signatures and the unchanged prefix link almost every persistent record. Insert/delete churn weakens Gram matching at low overlap, but prefix and clean residual-norm R@1 remain above 0.9996; micro-keys remove shared residual edges without closing those two channels.

The systems and privacy baselines add two less convenient but important findings. Real

CKKS/block-SIMD execution preserves ranking to numerical precision and cuts query upload by 74–87%, yet increases median latency in the tested TenSEAL layout because every candidate still returns a separate ciphertext. A formally calibrated Gaussian release, meanwhile, has no evaluated point that combines SHARD-matched utility with resistance to clean native-gallery linkage. In a strengthened GTR/Vec2Text outcome test, hiding the cell key lowers mean token-F1 from .665 to .242, but the wide public prefix and eight known pairs restore much of that loss. These results do not turn SHARD into DP or into a cryptographic document store; they make the boundary of the contribution measurable.

The global-linear baseline characterisation (alignment, PQ leakage, BEIR-truncation loss, and the calibrated-noise diagnostic that σ_{rec} is not a privacy metric) is retained in Section 7 as the foil that motivates SHARD, together with the CKKS ct-pt speed-up and parameter selection (established in our earlier work [56, 57]) that both schemes share. This global-linear stack is the lightweight hybrid of our earlier work [56]; SHARD is the attack-aware successor that removes its single weak axis.

Contributions relative to prior work. The global-linear baseline—SVD truncation, a secret rotation, a public ANN index and ct-pt CKKS reranking—together with the reproducible CKKS parameter selection are *not* new here: they are our previously published hybrid system [56] and its companion preprint [57], which we reproduce only as the comparison setting. Relative to that prior work, this paper begins with the SHARD cell-keyed residual split and its rank-preserving scoring correction (Section 6). It then follows the construction through the failure modes that determine its honest scope: anchor-span recovery below full key rank (Section 7.5); public-prefix, norm and within-cell leakage, including the micro-key limit and release churn (Sections 7.6 and 7.6); and an overlap-reference lookup (Section 7.7). The empirical case is completed by a two-encoder, ten-cell BEIR/MIRACL utility audit, an actual TenSEAL block-SIMD implementation, a formally accounted Gaussian-release baseline, and a learned text-level outcome. Everything else is reproduced baseline context, condensed and attributed accordingly.

Hypotheses. The study is organised around four testable hypotheses. **(H1)** Under diffuse disclosure, increasing C lowers the in-cell anchor rank and shifts partial-alignment recovery curves, without creating a hard full-rank threshold. **(H2)** The corrected full-dimensional split score preserves raw ranking, while half-PCA truncation may still lose graded retrieval quality. **(H3)** In the measured settings, the short prefix exposes less exact-neighbour structure than a public half-space PQ index, while shortlist recall is evaluated separately. **(H4)** Pairwise cross-key cosine is an insufficient unlinkability test because prefix, norm and Gram invariants can link re-keyed snapshots. The experiments of Section 7 test each in turn. These hypotheses concern the geometric construction; the CKKS, formal-DP and churn studies are complementary implementation and robustness audits.

Security-claim discipline. SHARD is an attack-aware *geometric* defence, not a cryptographic privacy guarantee, and every claim below should be read as holding *in the measured setting and against the studied attacker families*. The earlier baseline prototype uses CKKS to protect numerical query values and returned scores. Here SHARD also reuses that path in an actual TenSEAL implementation, so numerical error, serialization volume and phase latency are measured rather than inferred. This is still a single-process workstation benchmark, not a networked-service claim. The document side is geometrically transformed, not proven secure, and access patterns together with an overlapping public reference corpus remain exposed channels that we measure rather than close.

Notation. To avoid the overloading of the symbol N , we use N_{poly} for the CKKS polynomial-modulus degree, N_{docs} for the corpus size, and K_{cands} for the stage-1 short-list length throughout.

An elementary proxy bound. We state a one-line projection bound (Proposition 1): any decoder whose image is constrained to $\text{span}(V_k)$ has L_2 reconstruction error at least $\|\mathbf{x}_\perp\|_2$. This is a Pythagorean fact, not a security theorem; we use it only to make σ_{rec} a precise distortion proxy within that decoder class. Its empirical relation to the BLEU of an off-the-shelf VEC2TEXT attack is reported as Empirical Hypothesis 1.

Roadmap. Section 2 reviews related literature; Section 3 states the threat model; Section 4 describes the global-linear baseline (the foil) and the shared CKKS/ct-pt machinery; Section 5 an elementary projection bound; Section 6 introduces SHARD; Section 7 reports the baseline characterisation and the expanded SHARD evaluation: corrected utility, partial alignment, public-index leakage, measured CKKS, cross-release linkage with churn, a formal Gaussian release, learned text inversion, and reference lookup; Section 8 discusses limitations and open work. Appendix A condenses the baseline inversion analyses that overlap with the global-linear paper [57].

2 Related Work

Embedding inversion. VEC2TEXT [7] trains an encoder-decoder model that iteratively rewrites a candidate hypothesis until its embedding matches a target vector; on 32-token MS-MARCO inputs the attack achieves BLEU $\approx 97.3\%$ and $\approx 92\%$ exact recovery. GEIA [8] instead trains a generative decoder that maps a target embedding to a prompt and reconstructs the whole sentence in a single pass. TEIA [9] removes the need to query the victim encoder, training a surrogate inverter that transfers across embedding models. Earlier work of Song and Raghunathan [19] already established that embedding models leak substantial information about their inputs. Adjacent work on membership-inference [20] and on extracting training texts from large language models [21] confirms that the risk surface of dense retrievers is broader than the embedding-inversion literature alone suggests. The 2025 generation of attacks further weakens any defence that relies only on an unknown embedding orientation: ALGEN aligns a victim embedding space to an attack space with few known pairs and then generates text [10]; ZSINVERT removes the need for an embedding-specific inversion model by using a universal zero-shot decoding strategy [11]; and LAGO extends few-shot inversion to cross-lingual settings by coupling alignments across related languages [12]. These papers motivate the alignment experiment in Section 7.1: if a secret rotation can be absorbed by a small alignment matrix, it must not be described as a security guarantee.

Encrypted search and leakage. A long cryptographic line studies search over encrypted data and the leakage it admits by construction. Searchable symmetric encryption formalised the efficiency/leakage trade-off [51], and structured encryption generalised it to encrypted data structures with controlled disclosure [52]. A parallel line of leakage-abuse cryptanalysis shows that access-pattern and co-occurrence leakage alone can reconstruct queries and data [53, 54]. We position SHARD within this framing rather than against it: CKKS reranking hides the value channel, but the public stage-1 index and the per-query access pattern are leakage that remains *by construction*, which is why we measure the public-index neighbour leakage explicitly and flag access patterns as an open channel for composition with a PIR/ORAM-style primitive. A recent reproducibility study of VEC2TEXT [55] corroborates that inversion is strong under ideal conditions yet sensitive to sequence length and to simple quantisation/noise, consistent with our weak-attacker calibration.

Lightweight defences and transformed representations. Recent defences increasingly treat embeddings as representations whose geometry can be compressed, transformed or regularised rather than merely noised. EGUARD uses a projection network with mutual-information

Table 1: Positioning against recent privacy-preserving retrieval defenses. ✓ provided, × not addressed, ∼ partial/approximate. Rank fidelity refers to plaintext scoring; CKKS remains approximate.

Defense	Query priv.	Doc. priv.	Rank fidelity	Limited leak	Attacker / cost
SHARD (ours)	∼ CKKS path	∼ keyed	✓ plain.	×	prefix/norm/Gram; active cells
STEER [58]	✓ align.	×	∼ approx.	—	query inversion; one-off
ENTROGUARD [59]	×	∼ perturb.	×	∼	trained EIA; low
EGUARD [13]	×	∼ learned	×	∼	trained EIA; low
PQ-only / quantization	×	∼ lossy	×	∼	reconstruction; low
legacy uncalibrated Gaussian	×	∼ noise	×	×	distortion diagnostic; low
analytic Gaussian release [23]	×	✓ DP	×	∼	formal content DP; high utility cost

regularisation to reduce inversion leakage [13]; SPARSE studies concept-aware privacy mechanisms and anisotropic noise for embedding inversion resistance [14]; and nonparametric variational DP work adds bottleneck/noise mechanisms at the representation level [15]. Matryoshka-style representation learning shows that retrieval utility can be deliberately concentrated in prefix subspaces [16], while recent cross-space translation work argues that embedding spaces share enough geometry for transfer maps to be useful [17]. SHARD builds on both observations: it uses a Matryoshka-style prefix as a deliberately coarse public channel, precisely because the cross-space-translation literature shows that a single global geometry is alignable.

Positioning against the 2025 defense and attack lines. Two recent defenses target different channels than SHARD. STEER [58] is *query-side*: the client learns a one-off transform that maps its query into the server’s embedding space, so the server cannot invert the query—but it does not protect the stored documents and its cross-model alignment is approximate. ENTROGUARD [59] (like EGUARD) is a *perturbation* defense: it raises the entropy of inverted text under a distortion bound, protecting embeddings against trained inversion but at a lossy, non-exact reranking cost. On the attack side, ZSINVERT and the universal geometry of VEC2VEC make a leaked store inversion-prone without victim-specific training—precisely the alignment surface SHARD hardens. Table 1 places these lines on common axes. SHARD combines a rank-correct geometric split with a CKKS-compatible query path, but its document view is linkable through the prefix, norms and cell geometry. We therefore avoid an “only” or end-to-end privacy claim.

Template criteria and decoupled encoders. Cancellable biometrics [34] require renewability and unlinkability, criteria that orthogonal re-keying alone does not satisfy here: Experiments 25 and 28 link re-keyed records through invariant features, both with and without insert/delete churn. Work on retrieval with asymmetric document/query encoders [5, 6] shows that the document and query sides need not share one geometry—the premise that lets SHARD transform the stored side and the scoring query consistently. Our contribution is the resulting compartmentalisation and, equally, the measurement of where it fails the stronger template criteria.

Differential privacy for embeddings. Lyu et al. [25] investigate per-coordinate Gaussian noise on dense representations. Our formal replacement-adjacency baseline uses analytic Gaussian calibration and makes the operating-point tension concrete: at $\varepsilon = 1$, nDCG@10 is at most 0.011, while every strict match to corrected SHARD utility occurs at $\varepsilon = 32768$ with native-gallery linkage R@1 at least 0.995. Random-projection-based privacy [47, 27, 26] predates modern dense retrieval; the data-dependent SVD projection used here is closer to a noise-removal technique than to a privacy-preserving DP mechanism, although σ_{rec} provides a useful proxy quantity.

Homomorphic encryption. The CKKS scheme [28] introduced approximate-arithmetic FHE; OpenFHE [35] and TenSEAL [29] provide production-grade implementations. The bootstrapping cost [30] pushes practitioners to leveled circuits; specialised compilers [36, 37] optimise scale and

modulus chains. GPU acceleration of CKKS bootstrap [38] and Intel HEXL [39] have made larger leveled circuits feasible.

Hybrid privacy-preserving retrieval. Tiptoe [40] combines clustering with PIR for query privacy. SealPIR [41], OnionPIR [43] and SimplePIR [42] provide single-server PIR with practical query sizes. The architecture in this paper is complementary: ct-pt CKKS reranking takes care of the value side of the leakage, while access patterns remain a separate problem for which PIR/ORAM-like primitives are an appropriate composition.

3 Threat Model

The global-linear baseline and SHARD share one threat model, which we fix here. It is asymmetric—a *trusted client* (the data owner), an *honest-but-curious server*, and a *non-adaptive* baseline attacker, strengthened by the *known-plaintext* attacker we use throughout the alignment analysis. This is a fat-client, data-owner setting: the client runs the encoder, holds all keys, performs stage-1 retrieval locally, and the server only reranks under CKKS—a scoping assumption, not the only possible deployment. Tables 2 and 3 map the *baseline*’s coverage and claims (the foil); Table 4 does the same for SHARD (the contribution, Section 6).

The setting has three parties. The *client* is the data owner: it runs the encoder E , generates the secret keys $\mu, V_k, R, sk_{\text{CKKS}}$ and never shares them, so it is trusted both in software and in key management. The *server* runs the search service over the rotated database $E_{\text{rot}} \in \mathbb{R}^{N_{\text{docs}} \times k}$ and the public PQ artefact; it is honest-but-curious, following the protocol faithfully while trying to read textual information out of everything it stores and every query it processes. The *attacker*, finally, is at minimum non-adaptive—an off-the-shelf VEC2TEXT model with no knowledge of R —but against both schemes we grant it considerably more: a known-plaintext alignment observer (diffuse and targeted), public-index leakage, reference-corpus lookup, rank-deficient Procrustes, minimum-norm and regularised linear maps, and cross-release matching from prefix, norm and Gram invariants (Sections 7.5 and 7.6). Malicious-server behaviour remains out of scope; the learned text-inversion study is reported separately and does not turn the geometric tests into a universal adaptive-attacker guarantee.

Three distinct privacy notions. A recurring source of confusion in hybrid designs is the conflation of *what* is protected. We therefore separate three independent notions and tie each to the component responsible for it. *Document privacy* is the weakest: the plaintext rotated database E_{rot} is visible to the server, protected only by SVD truncation (through the proxy quantity σ_{rec}) and the secret rotation R , an empirical obfuscation layer—this is *not* a cryptographic protection, and it fails under a known R and, as Section 7.1 shows, under even a modest known-plaintext alignment attack. *Query-value confidentiality* comes from CKKS: the numerical scoring vector and returned scores are encrypted, and our measured SHARD path keeps the secret key client-side. It still reveals active cells, ciphertext counts and candidate identifiers. *Access-pattern privacy* is again absent—the candidate IDs reranked per query, and the public PQ codes, are not hidden, since CKKS leaves that channel untouched and closing it would need composition with a PIR/ORAM-style primitive.

In particular, the public PQ artefact (codebook + per-document codes, trained in the rotated space) is itself a lossy compressed view of E_{rot} . Section 7.1 quantifies this leakage on a controlled sample; the result is strong enough that public PQ codes must be treated as a meaningful side channel rather than as a harmless index artefact.

Table 2 summarises the coverage and Table 3 maps each claim to the evidence that supports it. The diagonal of the coverage table is intentional: ct-pt reranking takes care of the value channel, but the access-pattern channel (which document IDs are reranked) remains open, and

Table 2: Threat coverage of the global-linear *baseline* (the foil). SHARD’s coverage of the same vectors is in Table 4.

Attack vector	Status	Comment
Snapshot of E_{rot} without R (off-the-shelf Vec2Text)	Partial	Heuristic; empirically validated against non-adaptive Vec2Text.
Network interception, leakage of similarity scores	Yes	CKKS at tc128; sk never leaves the client.
Known R (known-rotation attacker)	No	Reduces to σ_{rec} from SVD truncation only.
Known-plaintext pairs ($\text{text}_i, E_{\text{rot},i}$)	No	Evaluated in Section 7.1; roughly k pairs recover the rotation numerically.
Reference-corpus overlap after known-plaintext alignment	Partial	Evaluated in Section 7.1; overlapping texts are recoverable.
Unsupervised cross-space translation attacks	No	Not evaluated on the baseline; a covariance control is not treated as a full translation attack.
Adaptive / learned alignment decoder	Partial	Experiment 29 applies the official pretrained GTR Vec2Text corrector to raw, SHARD and partially aligned views. A SHARD-specific trained or universal decoder remains unevaluated (Section 7.5).
Malicious server (protocol deviation, substitution)	No	Verifiable computation / TEEs are an orthogonal addition.
Access-pattern leakage on candidate IDs	No	CKKS hides values, not access patterns; requires PIR/ORAM.
Leakage from the public PQ artefact (rotated-space codes)	Partial	Evaluated in Section 7.1; neighbour structure is substantially preserved, so this remains an exposed channel.

the public-PQ-artefact channel is measured as leaky rather than closed. They would require composition with a PIR-style primitive and a stronger static-side index protection, respectively.

4 Baseline Construction (the global-linear foil)

This section describes the global-linear *baseline* that SHARD (Section 6) replaces—SVD truncation, a single secret rotation, a public PQ index, and CKKS reranking. The construction is *reproduced* from our prior work [56, 57] and condensed to the elements the SHARD comparison needs; we claim no novelty for it here and refer the reader there for the full derivation. It is the foil against which the contribution is measured; SHARD reuses its CKKS ct-pt machinery (Section 4.2) but discards its single global geometry. The baseline is illustrated in Figure 1: an offline data-preparation phase (Section 4.1) and the online query protocol (Section 4.2).

4.1 Offline phase

Encoder and centring. The data owner runs the encoder E on a corpus $D = \{T_1, \dots, T_{N_{\text{docs}}}\}$ to obtain $X \in \mathbb{R}^{N_{\text{docs}} \times d}$ with L_2 -normalised rows. The global centroid $\mu \in \mathbb{R}^d$ is computed and the corpus is centred: $X_c = X - \mu$.

SVD truncation. A randomised SVD [44, 46] $X_c \approx U_k \Sigma_k V_k^\top$ produces the orthonormal matrix $V_k \in \mathbb{R}^{d \times k}$ that spans the dominant k -dimensional subspace. We define the corpus-level relative reconstruction error

$$\sigma_{\text{rec}}^2(E; V_k) = \frac{\|E - \pi_k(E)\|_F^2}{\|E\|_F^2} = 1 - \eta_k,$$

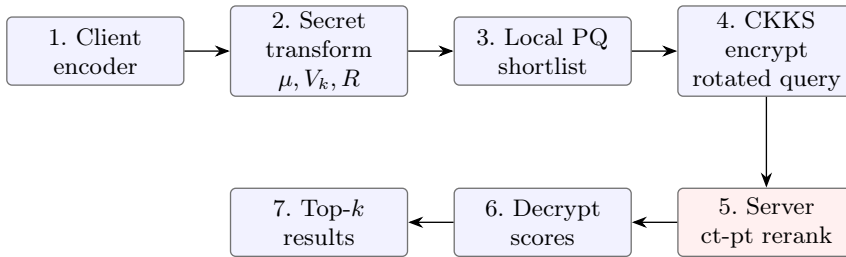
Table 3: *Baseline* claims vs. evidence (the foil): which statements about the global-linear stack are demonstrated and which are deliberately *not* claimed. SHARD’s claims are in Table 4.

Claim	Evidence / status
CKKS hides query values and scores from the server	Cryptographic construction; parameters from [57] (Sec. 4.3); query privacy is cryptographic.
ct-pt is faster than ct-ct for the reranking	Established in [57] ($1.7\times$ at the auto-tuned configuration); reused unchanged.
Secret rotation helps against <i>off-the-shelf</i> , non-adaptive Vec2Text	Appendix A; effect holds only for the tested weak-attacker configuration.
Aligned off-the-shelf Vec2Text recovers text/PII after Procrustes	Appendix A: not observed in the memory-capped RTX 4090 run, but the raw baseline is also near the reconstruction floor, so this is not a proof of security.
Secret rotation is robust to known-plaintext alignment	False ; Section 7.1 shows near-exact recovery with about k known pairs.
Public PQ codes are harmless metadata	False ; Section 7.1 shows substantial reconstruction and nearest-neighbour leakage.
The protective wrapper preserves ranking in $\text{span}(V_k)$	Metric-preserving, Pearson > 0.9999 [57] (recap Sec. 7.2).
SVD truncation improves $Acc@1$ (a “denoiser”)	Not significant [57]: paired tests give $p \geq 0.11$ for every encoder.
SVD truncation significantly improves $Acc@10$ for $d=1024$ retrieval-trained encoders	True [57] (e5-large +0.042, $p < 10^{-6}$); it significantly <i>degrades</i> compact/paraphrase encoders.
The denoiser generalises to real IR	False [57]: SVD significantly <i>reduces</i> nDCG@10 on BEIR SciFact/NFCorpus for every encoder.
σ_{rec} alone measures document privacy	Not claimed ; Appendix A shows that matched-distortion noise can preserve substantially more raw neighbour structure.
End-to-end query latency < 1 s at 10^6 docs	True [57]; recap Sec. 7.3 (loopback PoC).
The system is secure against an <i>adaptive</i> inversion attacker	Not claimed ; learned/universal adaptive decoders remain a required experiment (Sec. 9).
The system is secure against reference-corpus lookup	False when the reference overlaps ; Section 7.1 shows exact paragraph recovery after alignment.
The system is secure against unsupervised cross-space transfer attacks	Not claimed ; these attacks remain outside the evaluated threat model.
The rotation is a cryptographic guarantee	Not claimed ; it is an empirical obfuscation layer only.
Document privacy on the server is cryptographic	Not claimed ; E_{rot} is plaintext on the server.
No information leaks from the public PQ codes	Not claimed ; the measured leakage is non-negligible.

Table 4: SHARD (the contribution) claims vs. evidence.

Claim	Evidence / status
Corrected full-dimensional scoring preserves raw ranking	Supported algebraically and empirically; Experiment 23 obtains an identical order in every sampled float64 check across ten BEIR/MIRACL cells. End-to-end quality still depends on shortlist recall (Section 7.4).
Half-PCA alone causes the formerly reported 2–8 point loss	False; query centring caused part of it. Corrected truncation gaps range from approximately 0 to 5.3 nDCG points.
Cell keys create a hard d_{priv} recovery threshold	False; minimum-norm OLS exceeds 0.9 residual-gallery R@1 with about 32–36 in-cell pairs while the full key remains underdetermined (Section 7.5).
An unknown cell key suppresses off-the-shelf text reconstruction	Supported only in the strengthened GTR case; mean token-F1 falls from .665 to .242 without anchors, but the public prefix and eight-pair OLS restore substantial output. This is not a general inversion guarantee (Section 7.5).
Increasing C compartmentalises diffuse known pairs	Supported in the measured setting; the global OLS budget for $R@1 \geq 0.9$ shifts from 32 to 8192 over $C = 1$ to 256. The associated active-cell and packed-CKKS costs are measured separately; this is not a formal security threshold. Section 7.6; NN-overlap@10 0.20–0.55 vs. 0.76.
The public prefix leaks less than the baseline’s index	The first clause is supported; unlinkability is false. Norm-rank and the unchanged prefix link persistent rows almost perfectly even with insert/delete churn (Section 7.6).
Micro-keys remove shared residual edges and are unlinkable	False; it lacked adjacency, clipping and accounting. Experiment 27 replaces it with an analytically calibrated (ε, δ) Gaussian release under explicit replacement adjacency (Section 7.7).
The earlier Gaussian perturbation was a DP baseline	Supported for the local measured path; 315 real TenSEAL/SEAL runs cover encryption, serialization, ct-pt evaluation and decryption. Network RTT, concurrency and packed multi-score responses remain unmeasured (Section 7.4).
SHARD’s in-process cryptographic path is measured	False; Experiment 27 is a separately evaluated Gaussian-release baseline. No (ε, δ) guarantee is claimed for SHARD. Not claimed; same-cell similarities leak (Section 7.6).
SHARD itself is differentially private	False; the public prefix still de-anonymises (Section 7.7). Not claimed; it is a geometric compartmentalisation mechanism with explicit leakage.
SHARD hides the neighbour graph	
SHARD stops an overlapping reference lookup	
SHARD is a cryptographic guarantee	

Offline: protected database E_{rot} + public PQ artifact



Online: CKKS hides query values & scores; access pattern stays visible

Figure 1: Online query flow of the global-linear *baseline*. Steps 1–7 realise the two-stage protocol: client transformation $T(\cdot)$ and local PQ filtering produce a short-list of K_{cands} candidate IDs; step 4 encrypts the rotated query under CKKS; step 5 runs ct-pt reranking on the server; steps 6–7 decrypt the scores and sort. The secret keys $(\mu, V_k, R, sk_{\text{CKKS}})$ and the rotated database E_{rot} are produced offline by the data owner (Section 4.1).

where η_k is the fraction of squared Frobenius energy retained. Because σ_{rec} decreases as k grows, the operating point $k = d/2$ is the *largest* truncation dimension (rounded to $d/2$) at which the distortion floor $\sigma_{rec} \geq 0.10$ still holds uniformly across all five tested encoders (the binding cases are e5-large, $\sigma_{rec} = 0.101$, and mpnet, $\sigma_{rec} = 0.107$); larger k would drop some encoders below the floor. Thus $k = d/2$ maximises retained utility subject to the distortion-proxy constraint. We caution that the threshold 0.10 is an engineering proxy (Section 5), not a privacy guarantee, and is close to binding for two encoders.

Secret rotation. A Haar-uniform random orthogonal matrix $R \in O(k)$ is generated by QR-decomposing a Gaussian matrix and correcting the diagonal of R to obtain a uniform distribution on the group. The protected document representation is

$$v'_i = T(v_i) = RV_k^\top(v_i - \mu) \in \mathbb{R}^k.$$

Public PQ artefact. A faiss product-quantisation **IndexPQ** index [45] is trained *in the rotated space* E_{rot} with $M = k/4$ subquantisers and 8 bits per code. Both the codebook and the per-document codes are public; the client downloads them once at on-boarding. Training the PQ artefact in the rotated space removes one trivial cross-space pairing (an attacker cannot compose pairs $(\hat{E}_{proj}, E_{rot})$ from a non-rotated PQ index to estimate R). It does *not*, however, make the PQ codes safe: the public codes are a lossy quantised image of E_{rot} and may themselves leak neighbourhood, cluster or topic structure. We treat the PQ artefact as part of the attack surface and quantify nearest-neighbour and reconstruction leakage in Section 7.1.

Client-side state. The client retains μ , V_k , R and the CKKS secret key sk_{CKKS} . The server stores $E_{rot} \in \mathbb{R}^{N_{docs} \times k}$ in plaintext and the public PQ artefact.

4.2 Online query protocol

The pipeline of Figure 1 processes a query in stages, of which only the last leaves the client. The client encodes the query, $v_q = E(q_{text}) \in \mathbb{R}^d$, and forms two projected coordinates: the centred router $\tilde{q} = RV_k^\top(v_q - \mu)$ and the rank-correct scoring query $q_{score} = RV_k^\top v_q$. Using the public PQ artefact and $\tilde{q}$ it runs an asymmetric PQ-distance search locally and keeps the top- K_{cands} candidate IDs I_{cand} —a short-list small enough ($K_{cands} = 40$ in our experiments) that the CKKS reranking still fits the latency budget. It encrypts the scoring query, $ct_q = \text{Enc}_{pk}(q_{score})$, and sends it on; for each $j \in I_{cand}$ the server computes $ct_{score}^{(j)} = ct_q \odot v'_j$ in ct-pt mode, a ciphertext that encodes

$$\langle q_{score}, v'_j \rangle = v_q^\top V_k V_k^\top (v_j - \mu),$$

the projected raw score up to a document-independent query offset. The client finally decrypts those scores, $s_j = \text{Dec}_{sk}(ct_{score}^{(j)})$, sorts them, and returns the top- K documents—so the server reranks without ever seeing a plaintext query value or score.

Why ct-pt is sufficient. In step 5 the server multiplies a ciphertext by a plaintext vector. Unlike ciphertext-ciphertext multiplication, this operation produces a ciphertext that stays in the two-component form, no relinearisation is required, and the modulus chain is consumed by exactly one rescale. The sum of slots into a single scalar $\langle q_{score}, v'_j \rangle$ requires $\log_2 k$ rotations using Galois keys, but no further multiplications. The result is a sub-second latency at $N_{docs} = 10^6$.

Choice of K_{cands} . Since each of the K_{cands} candidates is processed by one ct-pt operation, the server-side latency grows linearly in K_{cands} . Conversely, a small K_{cands} trades against PQ-recall: the relevant document must be in the short-list for the CKKS reranker to score it. We pick K_{cands} as the smallest value for which $\text{PQ-recall}@K_{cands}$ matches the SVD-projected exact baseline

within the 5-seed CI; in our integral experiment this is $K_{\text{cands}} = 40$, but the parameter is a deployment knob and should be retuned for new encoders or larger corpora.

4.3 Cryptographic reproducibility

For the auto-selected configuration ($N_{\text{poly}} = 8192$, coefficient-modulus chain $[60, 40, 60]$ bits, $\log_2 Q = 160$, scale $\Delta = 2^{40}$, TenSEAL/Microsoft SEAL backend)—selected by the reproducible offline grid search of [56, 57] ($\approx 1.7\times$ faster than the TenSEAL stock $[60, 40, 40, 60]$ chain at equal security), reused here unchanged—we report the parameters needed to reproduce the cryptographic budget rather than only the latency:

A single scoring query $q_{\text{score}} \in \mathbb{R}^k$ packs into one ciphertext over $k \leq N_{\text{poly}}/2 = 4096$ slots, and since the integral encoders have $k \in \{192, 384, 512\}$ one ciphertext per query suffices, with no cross-ciphertext aggregation. Each candidate score then costs a single ct-pt multiply, one rescale (one level consumed), and a slot-sum realised as $\lceil \log_2 k \rceil$ ciphertext rotations under Galois keys (8–9 rotations for the tested k)—never a relinearisation or a ciphertext-ciphertext multiply. The $[60, 40, 60]$ chain leaves exactly one multiplicative level beyond the input, and after the lone rescale the remaining 60-bit modulus holds the additive noise far below the $\Delta = 2^{40}$ scale, so decrypted scores track the plaintext inner product to a relative error under 10^{-3} (Pearson > 0.9999 against exact, Section 7.2). The objects stay small: a fresh ciphertext is ≈ 0.21 MB, the Galois-key set for the power-of-two rotations is ≈ 6 –8 MB, the relinearisation key is *not* generated at all (ct-pt only), and the public key is ≈ 0.4 MB, so per query the client uploads one ciphertext (≈ 0.21 MB) and downloads the K_{cands} score ciphertexts ($\approx 40 \times 0.21 \approx 8.4$ MB before base64; measured on-wire JSON+base64 sizes are in Section 7.3). On security we assert no generic “standard” label, reporting instead $\log_2 Q = 160 < 218$ for $N_{\text{poly}} = 8192$, within the conservative ternary-secret **tc128** bound of the HomomorphicEncryption.org tables [31] and cross-checked against the lattice-estimator methodology of Albrecht et al. [32]; the standard documents are guidelines, so we name the concrete estimator and table used so the claim can be re-derived.

5 A Projection Bound for the Distortion Proxy

This short section concerns the *baseline* only (the bound is reproduced from [57]): it states the one elementary fact behind its σ_{rec} distortion proxy, and why that proxy is not a privacy metric—motivating SHARD’s move to an attack-aware design that does not rely on σ_{rec} at all. Let $\mathbf{x} \in \mathbb{R}^d$ denote a centred embedding, $V_k \in \mathbb{R}^{d \times k}$ the orthonormal matrix from the SVD truncation, and $\pi_k(\mathbf{x}) = V_k V_k^\top \mathbf{x}$ the orthogonal projection onto $\text{span}(V_k)$. We use a *per-vector* relative reconstruction error here, $\sigma_{\text{rec}}(\mathbf{x}; V_k) = \|\mathbf{x} - \pi_k(\mathbf{x})\|_2 / \|\mathbf{x}\|_2$, distinguished from the corpus-level Frobenius quantity $\sigma_{\text{rec}}(E; V_k)$ of Section 4.1 (the operating-point threshold 0.10 is applied to the corpus-level aggregate); write $\mathbf{x}_\perp = \mathbf{x} - \pi_k(\mathbf{x})$.

Proposition 1. *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be any decoder whose image is constrained by $f(\mathbf{y}) \in \text{span}(V_k)$ for every $\mathbf{y} \in \mathbb{R}^d$. Then for every $\mathbf{x} \in \mathbb{R}^d$*

$$\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 \geq \|\mathbf{x}_\perp\|_2^2,$$

with equality at $f(\mathbf{y}) = \mathbf{y}$. In relative form, $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2 / \|\mathbf{x}\|_2 \geq \sigma_{\text{rec}}(\mathbf{x}; V_k)$.

Proof. Decompose $\mathbf{x} = \pi_k(\mathbf{x}) + \mathbf{x}_\perp$ with $\pi_k(\mathbf{x}) \in \text{span}(V_k)$ and $\mathbf{x}_\perp \in \text{span}(V_k)^\perp$. By the image restriction $f(\pi_k(\mathbf{x})) \in \text{span}(V_k)$, hence $\mathbf{x} - f(\pi_k(\mathbf{x})) = (\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))) + \mathbf{x}_\perp$ is the sum of two orthogonal vectors. Pythagoras gives $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 = \|\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))\|_2^2 + \|\mathbf{x}_\perp\|_2^2 \geq \|\mathbf{x}_\perp\|_2^2$. $\square$

Scope: a projection bound, not an inversion-security theorem. Proposition 1 bounds the L_2 error of decoders *whose image is contained in* $\text{span}(V_k)$. It is a statement about information lost to the projection, not about the security of text inversion. A realistic attacker is not so

constrained: it may output arbitrary text, whose re-embedding generally has a non-zero $\mathbf{x}_\perp$ component, and it may exploit corpus priors or memorisation to partially recover the discarded component. The proposition therefore does *not* lower-bound the achievable inversion BLEU, token overlap or PII recovery. We use it only to make the *proxy* criterion $\sigma_{rec} \geq 0.10$ precise within the projection-restricted class, and we stress that the threshold 0.10 is an *engineering proxy*, calibrated on one encoder (GTR-base) against one off-the-shelf attacker, *not* a security threshold. A per-encoder ablation of σ_{rec} against BLEU, token overlap and typed-PII recovery (names, addresses, e-mail, phone, medical terms) is listed as a required pre-submission experiment (Section 9).

Empirical Hypothesis 1. *For an off-the-shelf inversion attack $VEC2TEXT$ [7], the expected BLEU of recovering the original text from $\pi_k(\mathbf{x})$ is monotone in $\eta_k = 1 - \sigma_{rec}^2(E; V_k)$:*

$$\mathbb{E}[\text{BLEU}(f(\pi_k(\mathbf{x})), T(\mathbf{x}))] \approx \text{BLEU}_0 + \gamma_f \eta_k,$$

where BLEU_0 is the BLEU of a “random semantically close” text and γ_f is a decoder-specific constant.

Hypothesis 1 is examined numerically in Appendix A; we deliberately separate the formal statement from the empirical observation.

6 SHARD: Cell-Keyed Residual Splitting

The baseline of Section 4 protects the collection with a single global geometry $E_{\text{rot}} = R V_k^\top (x - \mu)$. Its weakness is structural: one global map is recoverable by one alignment (Section 7.1), and the public index sits on the full protected space. SHARD keeps the same CKKS ct-pt reranking but removes the single global axis. It is a *family* of transforms parametrised by a cell count C .

Offline construction. Let μ be the corpus centroid and $V \in O(d)$ the PCA basis fitted to centred document embeddings (eigenvectors of the covariance, sorted by variance). For a document x_i write

$$V^\top (x_i - \mu) = [u_i \parallel r_i],$$

splitting its centred PCA coordinate into a *public prefix* $u \in \mathbb{R}^{d_{\text{pub}}}$ (the top variance directions) and a *private residual* $r \in \mathbb{R}^{d_{\text{priv}}}$, $d_{\text{priv}} = d - d_{\text{pub}}$. A coarse partition into C cells is defined by k -means on u . The data owner assigns one orthogonal key $H_c \in O(d_{\text{priv}})$ per cell. The experiments sample dense Haar-like matrices by Gaussian QR under deterministic research seeds; a deployment would need a CSPRNG-derived key schedule and should benchmark a structured fast-orthogonal representation. The stored representation of document i in cell $c(i)$ is the pair

$$(u_i, \quad z_i = H_{c(i)} r_i),$$

the prefix u_i for the stage-1 index and the keyed residual shard z_i for reranking. The keys $\{H_c\}$, V and μ are client secrets.

Online two-stage query. Routing and final scoring use two related query coordinates. The centred prefix $\tilde{u}_q = V_{\text{pub}}^\top (q - \mu)$ drives stage-1 ANN, while final scoring uses

$$u_q = V_{\text{pub}}^\top q, \quad r_q = V_{\text{priv}}^\top q.$$

The two prefix vectors differ only by the fixed offset $V_{\text{pub}}^\top \mu$. The client (i) runs stage-1 ANN locally with $\tilde{u}_q$ to obtain K_{cands} candidates; then (ii) for every active cell c forms $H_c r_q$. In the width-1 layout it sends one $\text{Enc}(H_c r_q)$ per cell; block-SIMD optionally concatenates B cell vectors

in one ciphertext, with zero-masked plaintext operands selecting the candidate’s block. The server returns ct-pt inner products against the stored shards z_i . Because H_c is orthogonal,

$$\langle H_c r_q, z_i \rangle = \langle H_c r_q, H_c r_i \rangle = \langle r_q, r_i \rangle,$$

in exact plaintext arithmetic. CKKS adds its usual numerical approximation. The scoring-prefix term can be added as plaintext to the encrypted response before decryption, as in Experiment 26, or combined by the client afterwards:

$$S(q, i) = \langle u_q, u_i \rangle + \langle r_q, r_i \rangle = q^\top (x_i - \mu) = q^\top x_i - q^\top \mu.$$

The last term is document-independent, so the exact-arithmetic ordering is the raw dot-product ordering. In the plaintext evaluation, the remaining measured retrieval loss is whether stage 1 places a relevant document in the short-list; Experiment 26 performs the complementary check with actual CKKS: across 315 trials it finds no top-1 flips, unit top-10 overlap and maximum absolute score error 2.29×10^{-6} .

Proposition 2 (Rank-preserving cell-keyed reranking). *Let $H_c \in O(d_{\text{priv}})$ be the orthogonal key of cell c and $z_i = H_{c(i)} r_i$ the stored shard of document i in cell $c(i)$. For any query residual r_q and any document i ,*

$$\langle H_{c(i)} r_q, z_i \rangle = \langle r_q, r_i \rangle,$$

so $S(q, i) = q^\top x_i - q^\top \mu$ is reconstructed with no truncation in exact arithmetic. Consequently, $S(q, i) - S(q, j) = q^\top x_i - q^\top x_j$, and the raw document ranking is preserved. CKKS evaluation is approximate, not algebraically exact. Consequently, for documents i, j in the same cell, $\langle z_i, z_j \rangle = \langle r_i, r_j \rangle$.

Proof. H_c is orthogonal, hence $H_c^\top H_c = I$ and $\langle H_c r_q, H_c r_i \rangle = r_q^\top H_c^\top H_c r_i = \langle r_q, r_i \rangle$; the prefix and residual terms together give $q^\top (x_i - \mu)$. Its offset from $q^\top x_i$ is constant across documents. The same-cell identity applies the same equality to the shared key H_c of two stored residuals. $\square$

Cross-release invariants and a non-claim. Orthogonal keying is invertible for a party that holds the key; without the key it is an unidentified orientation, not a non-invertible template. Fresh orientations also do not make two snapshots unlinkable. For two releases a, b of the same residual under independent orthogonal keys,

$$\|z_i^{(a)}\|_2 = \|r_i\|_2 = \|z_i^{(b)}\|_2, \quad \langle z_i^{(a)}, z_j^{(a)} \rangle = \langle r_i, r_j \rangle = \langle z_i^{(b)}, z_j^{(b)} \rangle$$

whenever i and j share a cell. The prefix u_i is unchanged as well. Norms, within-cell Gram signatures and the prefix are therefore stable cross-release identifiers. Re-keying changes coordinates and preserves plaintext retrieval, but SHARD does not claim unlinkability, irreversibility or a cancellable-template guarantee. We measure these channels explicitly in Section 7.6.

Proved vs. measured, and what SHARD does not promise. What SHARD *proves* is precisely Proposition 2: raw-rank-equivalent full-dimensional reranking in plaintext algebra, and the same-cell cancellation $\langle z_i, z_j \rangle = \langle r_i, r_j \rangle$. Everything else below is *measured*, not guaranteed: partial known-pair alignment (§7.5), learned and unsupervised attackers (§7.5), public-prefix leakage (§7.6), micro-key and cross-release behaviour (§7.6), and online cost (§7.4). Three non-promises follow directly from the proposition and the construction: SHARD is *not* a cryptographic guarantee on the store; it does *not* hide the coarse neighbour graph (same-cell similarities are computable by the server unless $C=N$); and an *overlapping plaintext reference corpus* still de-anonymises through the public prefix (§7.7). The effect SHARD measures is compartmentalisation of a *diffuse* alignment stream, not a hard privacy barrier or protection of the coarse graph.

The C family and micro-keys. The cell count interpolates between two regimes. At $C=1$ the residual carries a single global key; this is a full-dimensional SHARD control, not the earlier truncated baseline. At the other extreme, a *micro-key* variant assigns one key per document ($C=N$). No two residuals then share a key, so the directly comparable within-release residual graph vanishes, at the cost of one transformed query per candidate. This does not remove the stable prefix or residual norm and therefore does not imply cross-release unlinkability.

What the server sees, and what it protects. The honest-but-curious server holds the coarse prefix u , cell labels and the cell-keyed shards z ; it also observes candidate identifiers, ciphertext counts and the query access pattern. Three facts frame every experiment below. (1) *Within a cell the keys cancel:* $\langle z_i, z_j \rangle = \langle r_i, r_j \rangle$ for i, j in the same cell, so same-cell similarities are computable by the server; SHARD localises but does not eliminate neighbour-graph leakage. (2) *Partial mapping precedes exact key recovery:* known pairs identify the key action on their span even while the complete H_c remains underdetermined. Exact key identification generically needs full in-cell rank, but retrieval and record matching may succeed much earlier. (3) The stable prefix, residual norms and within-cell Gram signatures provide direct cross-release linkage channels. Cell sharding reduces the rate at which a diffuse anchor stream fills any one cell; it does not create a hard privacy threshold.

7 Experiments

The experiments come in two groups. Sections 7.1–7.3 characterise the global-linear baseline, opening with the failure modes that motivate SHARD: known-plaintext Procrustes alignment recovers the rotation from $\approx k$ pairs, the public PQ codes leak neighbour structure, and an overlapping reference corpus turns alignment into exact lookup (Section 7.1); the baseline’s utility and latency are then recapped from [56, 57]. The inversion-versus-rotation analyses reproduced from the global-linear paper [57]—off-the-shelf and aligned Vec2Text, the lightweight projection baselines, and the calibrated-noise diagnostic that σ_{rec} is not a privacy metric—are condensed in Appendix A. Sections 7.4–7.7 then *evaluate SHARD*: utility, the online cost, alignment under rank-deficient and minimum-norm estimators, public-index leakage, cross-release linkage for cell and micro-keys, and the overlap-reference limitation. The maximal audit then adds a real CKKS/block-SIMD execution, insert/delete churn, a formally calibrated Gaussian release, and a learned text-inversion outcome. The former uncalibrated “DP-noise” comparison is retained only as an artifact-side distortion diagnostic. Readers interested only in the contribution can skip to Section 7.4; the baseline group is the foil.

Hardware and software. All local experiments run on a single workstation: Intel Core i5-14400F, 32 GB DDR5-4800, NVIDIA RTX 5060 (8 GB VRAM), Windows 11. The baseline artifact retains its original pinned environment and CKKS build information. The corrective Experiments 23–25 reported here ran under Python 3.12.3, NumPy 2.4.6, SciPy 1.17.1, scikit-learn 1.8.0 and Matplotlib 3.10.9; each output directory records the command, seeds, configuration and Git revision. Experiments 26–28 use the same workstation. Experiment 26 adds TenSEAL 0.3.16 and psutil 7.2.2, pins one logical thread per P-core, and executes SEAL CKKS on the CPU; Experiments 27 and 28 use the recorded NumPy/SciPy stack. The local learned-inversion run uses the RTX 5060 and its own frozen package report. The earlier aligned Vec2Text stress test retained in Appendix A used a separate memory-capped RTX 4090 profile with roughly 11 GiB free VRAM, Python 3.12 and PyTorch 2.6.0+cu124. We keep these environments distinct rather than treating latency across machines as comparable.

Table 5: Known-plaintext Procrustes attack. $m = -1$ is the known- R oracle; $m = 0$ is no alignment. Values are means over five rotation seeds.

Known pairs m	Mean cosine	Rel. L_2 error	Target R@1	Target R@10
0	-0.004	1.417	0.000	0.001
10	0.098	1.343	0.005	0.039
50	0.377	1.116	0.722	0.925
100	0.621	0.870	1.000	1.000
192	1.000	1.8×10^{-5}	1.000	1.000
500	1.000	4.9×10^{-7}	1.000	1.000

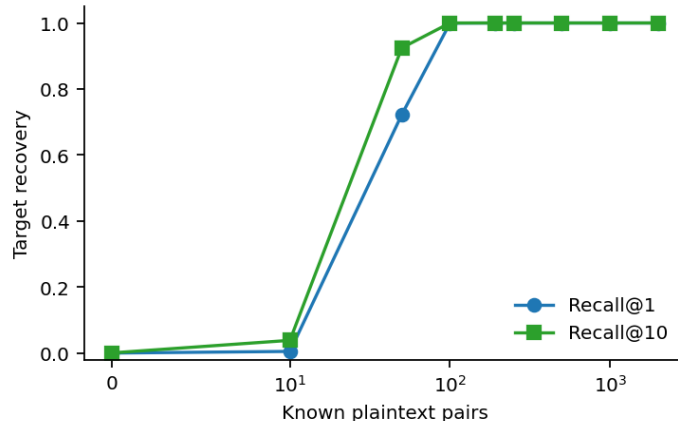


Figure 2: Known-plaintext alignment quickly destroys the unknown-rotation assumption. The x axis uses a symlog scale to include $m = 0$.

7.1 Attacks on the baseline: alignment, PQ leakage, and inversion

We now stress the global-linear baseline with four attacks that step outside the primary non-adaptive threat model: known-plaintext alignment of the secret rotation, leakage from the public PQ codes, a text-level reference-corpus lookup, and an aligned off-the-shelf inverter.

Known-plaintext alignment. We simulate an attacker who knows m pairs $(E(x_i), E_{\text{rot},i})$ and estimates the hidden orientation with the orthogonal Procrustes solution [33]. The experiment uses e5-small projected to $k = 192$, five rotation seeds, 400 held-out probes and a 10 000-document gallery. The attack is evaluated in embedding space: after alignment, the probe should recover its own projected vector and retrieve the corresponding document from the gallery. This does not measure generated text quality, but it is the relevant precursor to a few-shot Vec2Text-style attack because it recovers the native projected space given to the inverter.

The conclusion is sharp: the rotation is useful only while aligned plaintext anchors are unavailable. With approximately k known pairs, the orientation is numerically recovered. Even 50 pairs do not recover the vector perfectly, but they are enough to retrieve the target in the top 10 for 92.5% of probes in a 10 000-document gallery. This is a negative result for broad document-privacy claims and is therefore reflected in the threat table.

PQ-code leakage. We also train public product-quantisation codebooks directly in the rotated space and reconstruct approximate vectors from codebook + per-document codes alone. On a 20 000-document sample, the canonical e5-small configuration ($M = 48$, 8 bits) preserves a mean cosine of 0.953 to E_{rot} , 67.4% of exact top-10 neighbours, and 97.1% of exact top-10 neighbours inside the approximate top-40 candidate set. Smaller artefacts leak less but remain non-negligible (Table 6).

Table 6: Leakage from public PQ codes on a 20 000-document rotated e5-small sample. Neighbour overlap excludes the query document itself.

M	Bits	Bytes/vector	Cosine to E_{rot}	NN overlap@10
24	8	24	0.837	0.384
48	8	48	0.953	0.674
48	6	36	0.904	0.529

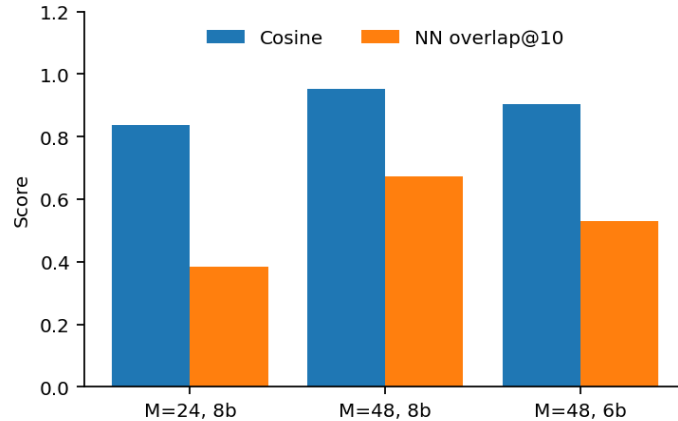


Figure 3: Public PQ codes preserve both vector direction and substantial nearest-neighbour structure in the rotated space.

Table 7: Reference-corpus lookup attack after known-plaintext alignment. The overlapping reference has 500 targets plus 100 000 decoys. Jaccard is token-set overlap between the target paragraph and the top-1 retrieved paragraph. Values are means over five rotation seeds.

Known pairs m	Exact R@1	Exact R@10	Jaccard overlap-ref	Jaccard disjoint-ref
0	0.000	0.000	0.006	0.006
10	0.000	0.009	0.009	0.009
25	0.044	0.132	0.055	0.012
50	0.535	0.784	0.543	0.018
100	0.998	1.000	0.999	0.034
192	1.000	1.000	1.000	0.043

These results do not invalidate the query-privacy contribution of CKKS, but they do narrow the document-privacy interpretation: public PQ artefacts and known plaintext anchors should be treated as exposed channels. A system requiring stronger document confidentiality needs a different static-side primitive or a composition that hides the PQ artefact itself.

Reference-corpus lookup after alignment. The Procrustes experiment above is an embedding-space test. We next ask whether the same failure mode can become a text-level leakage channel without training a generative inverter. The attacker is given the protected target vectors, m known plaintext/protected pairs for estimating the rotation, and a reference corpus of candidate paragraphs with native e5-small embeddings. This models a realistic overlap case: some private documents may also occur in a public or previously leaked corpus. We use 500 target paragraphs, 100 000 reference decoys, five rotation seeds and the same $k = 192$ SVD subspace. The overlapping reference corpus contains the targets plus the decoys; the disjoint reference corpus removes the targets and is used only as a lexical nearest-neighbour proxy.

The overlap case is severe. With only 50 known pairs, the attacker recovers the exact source paragraph at 53.5% top-1 and 78.4% top-10 recall among 100 500 candidates; with 100 known pairs,

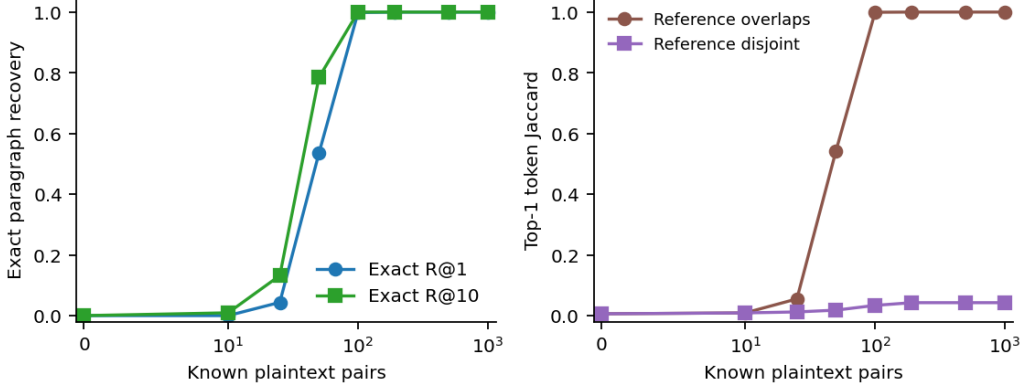


Figure 4: Reference-corpus lookup after known-plaintext alignment. When the reference corpus overlaps the protected collection, alignment turns the protected vector into an exact paragraph lookup. With a disjoint reference corpus, this simple token-overlap proxy remains low; semantic leakage there requires a stronger judge or generative attack.

top-1 recall rises to 99.8%. This is the text-level analogue of the embedding-space Procrustes result and makes the static-side limitation concrete: if a public or leaked reference corpus overlaps the private collection, rotation secrecy is not enough. The disjoint-reference numbers are intentionally reported as a boundary condition: token Jaccard remains low (0.034 at $m = 100$ and 0.043 at $m = 192$), so a non-overlapping reference corpus requires semantic similarity metrics, human judgement, or a generative inverter before one can claim text reconstruction.

Aligned Vec2Text inversion stress test. As a portable generative-inversion control on the baseline, we also ran an aligned off-the-shelf VEC2TEXT corrector (GTR-base, $k = d/2$, up to $m = 500$ Procrustes pairs) against the rotated store. No exact document or typed PII is recovered in any case, but the *raw* row is also at the reconstruction floor, so this off-the-shelf corrector certifies nothing; the open risk is a learned, corpus-adapted decoder. The full setup and table are reproduced from [57] in Appendix A; the result does not contradict the exact-lookup failure mode of Section 7.1.

7.2 Baseline utility (recap)

The utility of the global-linear baseline is established in our prior work [56, 57] and is not reproduced here. In brief, at $k = d/2$ on the 10⁶-document corpus the PQ+CKKS wrapper is metric-preserving in $\text{span}(V_k)$ (CKKS-decrypted versus exact scores correlate at Pearson > 0.9999); the SVD step is a mild, encoder-dependent denoiser whose apparent Acc@1 gains are *not* statistically significant and which *significantly reduces* nDCG@10 on graded-relevance BEIR, so k is a tunable accuracy cost rather than a free denoiser, met within the end-to-end accuracy budget only for retrieval-trained $d \geq 768$ encoders. SHARD reuses the baseline’s ct-pt arithmetic but not its half-space score: in exact plaintext algebra it reranks in the full space (Section 6). Its own retrieval quality is evaluated in Section 7.4.

7.3 Baseline latency (recap)

End-to-end latency of the global-linear baseline is reported in full in [57]; the point that matters for SHARD is architectural. The pipeline reranks only the $K_{\text{cands}}=40$ stage-1 candidates, not 10⁶, giving a server-side p_{95} of 219–356 ms and ≈ 578 ms end-to-end once HTTP/JSON framing is added; the local PQ stage—not CKKS—reduces 10⁶ documents to 40, so the cryptographic cost is independent of corpus size N_{docs} . SHARD adds a two-stage rescoring whose marginal online cost is measured in Section 7.4.

Table 8: SHARD utility on three BEIR datasets (SciFact, NFCorpus, ArguAna; nDCG@10, $K_{\text{cands}}=200$). Each protected column reports the change Δ vs. raw with a paired-bootstrap 95% CI over per-query nDCG@10 (10^4 resamples, seed 2026). “Centred full” isolates the legacy query-centring error; “corrected half” isolates genuine half-PCA truncation; SHARD uses corrected full scoring after a centred $d/4$ router.

Encoder	Dataset	raw	centred full Δ [95% CI]	corrected half Δ [95% CI]	SHARD $d/4$ Δ [95% CI]
e5-base	SciFact	0.637	-0.051 [-.072, -.031]	-0.015 [-.028, -.002]	+0.001 [.000, .003]
e5-base	NFCorpus	0.327	-0.022 [-.031, -.012]	0.000 [-.005, .005]	-0.001 [-.002, .000]
e5-base	ArguAna	0.346	-0.080 [-.089, -.071]	-0.008 [-.012, -.003]	-0.001 [-.002, .000]
e5-small	SciFact	0.598	-0.039 [-.059, -.020]	-0.049 [-.068, -.030]	-0.008 [-.020, .001]
e5-small	NFCorpus	0.302	-0.007 [-.016, .002]	-0.012 [-.019, -.005]	-0.002 [-.004, .000]
e5-small	ArguAna	0.358	-0.038 [-.046, -.030]	-0.030 [-.038, -.022]	-0.002 [-.004, .000]

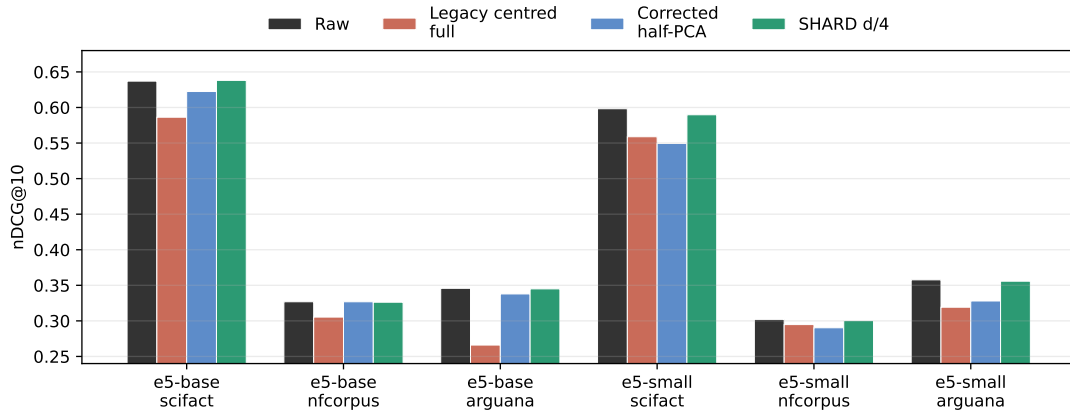


Figure 5: BEIR utility after correcting query centring. The corrected half-PCA gap is smaller than the legacy result but remains dataset-dependent; the $d/4$ router followed by corrected full scoring stays close to raw.

7.4 SHARD: utility and online cost

The experiments above characterise the global-linear baseline. We now evaluate SHARD on the same cached embeddings, computing the geometry with `numpy`. This verifies the plaintext identity; it is not a substitute for measuring CKKS approximation in an end-to-end prototype.

The corrected protocol exposes an error in the earlier comparison. Documents use $V^\top(x - \mu)$, but a ranking-preserving scoring query must use $V^\top q$. Centring both sides adds a document-dependent term and changes the ranking. Experiment 23 evaluates raw, legacy-centred, corrected half-PCA, and corrected two-stage scores on every judged query in six BEIR [50] and four MIRACL cells, with 10^4 paired-bootstrap resamples. In an independent float64 check the corrected full score and raw score have the same complete ordering in all tested queries (maximum numerical error 7.71×10^{-16}). Table 8 shows that the $d/4$ SHARD shortlist followed by corrected full reranking remains within 0.009 nDCG of raw; its small residual delta is shortlist recall, not score distortion.

Why this is not merely “do not truncate”. A reader might object that the baseline could also rerank full-dimensionally by storing the untruncated E_{rot} . It cannot do so for free: an untruncated plaintext store has a *single global key that cancels for all pairs*, so the server recovers the entire d -dimensional neighbour graph. The point of SHARD is narrower: full-dimensional reranking need not create a *globally* comparable residual geometry. It does create exact within-cell geometry and stable norm signatures, so it is not privacy-neutral. The utility result and the leakage result must be read together.

The corrected half-PCA result changes the interpretation. Truncation remains significantly

Table 9: SHARD utility on MIRACL (multilingual, nDCG@10 , $K_{\text{cands}}=200$): full-corpus retrieval over the dev queries of two typologically distinct languages. Columns as in Table 8: Δ vs. raw with a paired-bootstrap 95% CI over per-query nDCG@10 (10^4 resamples, seed 2026). $|C|$ is the corpus size; MIRACL uses binary judgements scored with the graded nDCG@10 metric.

Encoder	Language ($ C $)	raw	centred full Δ [95% CI]	corrected half Δ [95% CI]	SHARD $d/4$ Δ [95% CI]
e5-base	Swahili (132k)	0.707	-0.019 [-.032, -.007]	-0.012 [-.020, -.003]	+0.000 [.000, .001]
e5-base	Bengali (297k)	0.727	-0.031 [-.043, -.020]	-0.016 [-.025, -.007]	0.000 [.000, .000]
e5-small	Swahili (132k)	0.680	-0.036 [-.051, -.021]	-0.053 [-.068, -.037]	-0.006 [-.015, .001]
e5-small	Bengali (297k)	0.688	-0.033 [-.046, -.020]	-0.039 [-.055, -.024]	-0.009 [-.018, -.001]

Table 10: Structural active-cell cost (e5-small, $d_{\text{pub}} = d/4$). The last two columns retain the earlier one-0.21 MB-ciphertext model for comparison; Table 11 reports actual serialized traffic.

K_{cands}	active cells (mean / p95)		upload (MB)	$\times$ baseline
	$C=64$	$C=256$	($C=256$)	($C=256$)
40	7.2/13	11.6/19	2.4	11.6 $\times$
100	11.3/19	20.0/32	4.2	20.0 $\times$
200	15.7/26	30.1/48	6.3	30.1 $\times$

harmful on five cells, but e5-base/NFCorpus is effectively unchanged ($\Delta = -0.00002$); much of the previous baseline gap came from query centring rather than discarded dimensions. The SHARD $d/4$ deltas stay between -0.008 and $+0.001$, confirming the rank-preserving score and showing the remaining shortlist effect without overstating truncation.

Multilingual replication at 10^5 scale (MIRACL). To test whether the BEIR pattern holds on a large, *multilingual* benchmark scored with a graded ranking metric, we repeat the full-corpus protocol on MIRACL—the canonical multilingual IR benchmark (binary judgements scored with nDCG@10)—for two typologically distinct languages with manageable corpora: Swahili (Bantu, Latin script; 131,924 passages, 482 dev queries) and Bengali (Indo-Aryan, Bengali script; 297,265 passages, 411 queries). Table 9 shows the same pattern at 10^5 -document scale. Corrected half-PCA loses 0.012–0.016 nDCG on e5-base and 0.039–0.053 on e5-small; the legacy centring error is separately visible in every row. The corrected $d/4$ SHARD pipeline is equal to raw on e5-base and within 0.006–0.009 on e5-small. This reinforces $d/4$ as a reasonable router, while attributing each loss to the right source.

Online cost and the C trade-off. SHARD is not free relative to the single-query baseline, and we first expose the structural active-cell cost before measuring the concrete CKKS path. The client sends one encrypted residual query *per active cell*—a cell touched by the stage-1 short-list—so the per-query upload and client-side encryption scale with the number of distinct cells the short-list spans. Table 10 reports the active-cell counts. At $C=256$, $K_{\text{cands}}=200$ the short-list spans about 30 cells. The historical 6 MB figure obtained by multiplying this count by the baseline ciphertext size is deliberately kept as a first-order model; the measured serialization below supersedes it for the implemented layouts. The structural trade-off remains: larger C slows the accumulation of in-cell alignment evidence but costs more active-cell query transforms. The numbers are consistent across encoders (e5-base spans ≈ 9 –35 cells over the same grid). Active cells grow because k -means cells are finer than a query’s neighbourhood; a deployment tunes $(C, K_{\text{cands}}, d_{\text{pub}})$ to balance alignment resistance, stage-1 recall and upload, and the micro-key extreme ($C=N$) is the endpoint where every candidate is its own cell ($\approx K_{\text{cands}}$ queries).

Measured CKKS and block-SIMD packing. Experiment 26 implements the residual reranker with real CKKS operations in TenSEAL/SEAL [29], using cached multilingual-e5 SciFact

Table 11: Actual CKKS reranking at $K_{\text{cands}} = 128$ (15 trials per row). Traffic is the median serialized decimal MB; latency is the in-process online-path p50/p95. “ct” is the median number of uploaded query ciphertexts.

Encoder	pack B	query ct	upload	response	p50 (s)	p95 (s)
e5-small	1	30	10.03	30.13	2.415	2.504
e5-small	8	4	1.34	30.13	3.049	3.180
e5-base	1	31	10.36	30.13	2.655	2.741
e5-base	4	8	2.67	30.13	3.040	3.136

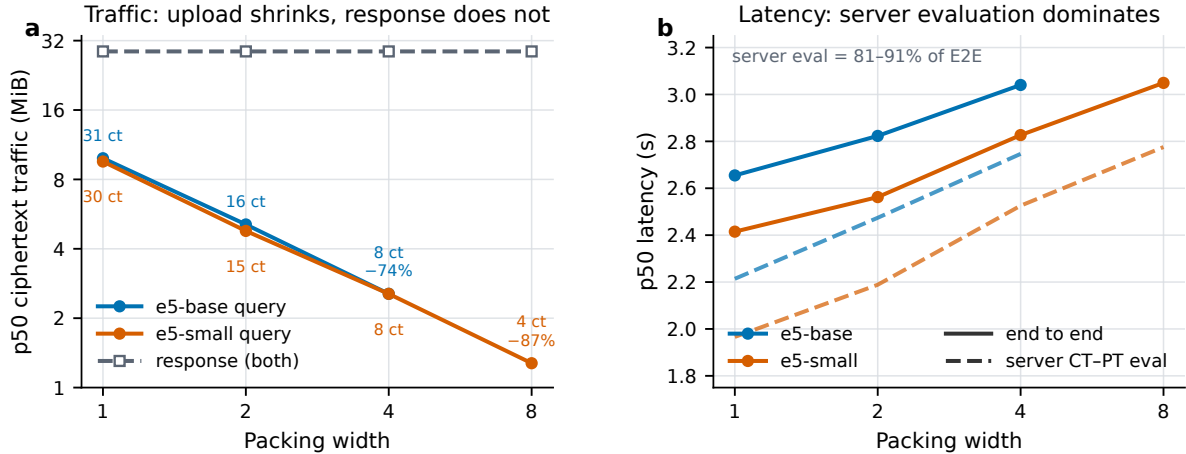


Figure 6: Measured block-SIMD trade-off at $K = 128$. Packing cuts query upload by 74.2% (e5-base, width 4) and 86.7% (e5-small, width 8), but the separate encrypted response per candidate remains 30.13 decimal MB (28.7 MiB on the axis). TenSEAL’s dot reduction over the larger logical vector increases p50 latency by 14.5% and 26.3%, respectively.

document/query embeddings for both encoder sizes. It is a systems benchmark on this real embedding geometry, not a claim of ten-dataset CKKS coverage. The public server context contains the public and Galois keys but no secret key. We use $N_{\text{poly}} = 8192$, a $[60, 40, 40, 60]$ coefficient-modulus chain and scale 2^{40} ; the provisioned server context is 34.08 MB and is excluded from per-query traffic. A width B layout concatenates B active-cell query blocks in one ciphertext. For a candidate in block b , the plaintext operand is zero outside that block, so one real ciphertext–plaintext dot product returns its residual contribution. The implementation adds the public-prefix score as a plaintext scalar to that encrypted response. The client then decrypts one encrypted full score per candidate.

The benchmark contains 315 measured trials over two encoders, $K_{\text{cands}} \in \{16, 64, 128\}$, three seeds, five repetitions and every layout that fits the 4096 CKKS slots. Each timing includes client transform, packing, encryption and serialization; server deserialization, plaintext-mask construction, ct–pt evaluation and response serialization; and client deserialization and decryption. Table 11 gives the most demanding $K = 128$ cells. Across the whole grid the maximum score error is 2.29×10^{-6} , there are no top-1 flips, top-10 overlap never falls below 1.000, and the minimum Kendall τ is 0.999754. The largest p95 point-sampled RSS increment is 46.8 MB (e5-base, $K = 128$, width 1); this is an approximate sampled increment rather than a high-frequency peak trace.

This result answers the implementation question more sharply than a byte model could. Block-SIMD packing is implemented and materially reduces upload and client encryption, but it is not a latency win in this layout: server evaluation occupies 81–91% of end-to-end time, while one response ciphertext is still emitted for every candidate. Packed multi-score output, real network RTT/TLS, concurrent clients and an ANN service are therefore systems work still to be

done; they are not silently folded into these local numbers.

Choosing C from measured recovery curves. The earlier draft derived a supposedly secure cell count by assuming that an attack fails until a cell contains d_{priv} anchors. Experiment 24 rejects that premise: partial estimators exploit a lower-rank anchor span. There is therefore no bandwidth-optimal secure C in this model. A deployment can only choose an empirical operating point by comparing the active-cell cost in Table 10 with an acceptable recovery curve under a specified disclosure budget. Increasing C dilutes a diffuse anchor stream; it does not stop an attacker who can place anchors in one target cell.

7.5 SHARD under partial alignment

We repeat the residual-only known-pair experiment without the artificial full-rank gate. An observer receives m nested pairs (r_i, z_i) , fits a reverse map in every target cell containing at least one pair, and matches 256 held-out targets against a 10 000-document native residual gallery. The e5-small pool has 100 000 documents, $d_{\text{pub}} = 96$ and $d_{\text{priv}} = 288$; we evaluate $C \in \{1, 16, 64, 256\}$ over three key/disclosure seeds. Confidence intervals use a hierarchical seed-then-target bootstrap with 500 repetitions.

The methods are rank-deficient orthogonal Procrustes, a Moore–Penrose minimum-norm linear map, ridge with both zero and cross-validated regularisation, and the polar projection of the linear map. All methods run for every positive in-cell anchor count. This is a stronger and fairer game than declaring an untouched keyed vector whenever the anchor matrix is not full rank.

Proposition 3 (Identifiable anchor span). *Let $z_j = Hr_j$ for an unknown $H \in O(d_{\text{priv}})$ and let $\mathcal{S} = \text{span}\{r_1, \dots, r_m\}$ have rank s . The pairs determine the action of H on $\mathcal{S}$, but leave an arbitrary orthogonal extension on $\mathcal{S}^\perp$. Thus exact identification of a generic full key requires $s = d_{\text{priv}}$, whereas the minimum-norm reverse map applied to a new target recovers its projection $P_{\mathcal{S}}r$. For an isotropic target,*

$$\frac{\mathbb{E}\|P_{\mathcal{S}}r\|_2^2}{\mathbb{E}\|r\|_2^2} = \frac{s}{d_{\text{priv}}}.$$

Proof. The paired vectors define an isometry between $\mathcal{S}$ and HS . Any two orthogonal maps that agree on $\mathcal{S}$ and differ only on $\mathcal{S}^\perp$ produce the same observed pairs. Writing the row matrices as $Z = RH^\top$, the minimum-norm least-squares reverse map is $Z^+R = HP_{\mathcal{S}}$, hence a new row $z^\top = r^\top H^\top$ maps to $r^\top P_{\mathcal{S}}$. The isotropic expectation is the trace identity $\mathbb{E}[r^\top P_{\mathcal{S}}r]/\mathbb{E}[r^\top r] = s/d$. $\square$

The proposition separates two questions that the previous analysis conflated. A complete key remains underdetermined below full rank, but record matching may need only a discriminative projection. Table 12 makes the distinction stark. Minimum-norm OLS exceeds 0.9 R@1 with only about 32–36 anchors in the average target cell, while full-rank coverage is zero in every row. The required *global* budget still grows approximately with C under diffuse disclosure (32, 512, 2048, 8192), so compartmentalisation is real; its absolute security interpretation is much weaker than the former d_{priv} -anchors-per-cell claim.

Attacker comparison and targeted disclosure. Ridge with $\alpha = 0$ is numerically identical to the Moore–Penrose result; cross-validated ridge differs by less than 2×10^{-6} . Orthogonal Procrustes and the polar projection are weaker below full rank but still reach R@1 above 0.9 at roughly 128 in-cell anchors: global $m = 128, 2\,048, 8\,192, 32\,768$ for $C = 1, 16, 64, 256$, respectively, again with zero full-rank coverage at the threshold. The earlier fixed Ridge($\alpha = 1$) comparison therefore did not establish that Procrustes was the strongest observer, and that claim is withdrawn. Likewise, a covariance-matching control is not a full implementation of an unsupervised translation attack.

Table 12: Strongest partial-alignment result (minimum-norm OLS, e5-small). The reported global m is the first tested point with mean R@1 at least 0.9. CIs are hierarchical 95% intervals. Exact full-key rank is absent at every operating point even though record matching is already reliable.

C	global m	mean anchors/cell	R@1 [95% CI]	recovered cosine	full-rank coverage
1	32	32.0	0.988 [0.979, 0.996]	0.356	0.000
16	512	35.8	0.939 [0.921, 0.957]	0.394	0.000
64	2048	34.8	0.953 [0.932, 0.970]	0.412	0.000
256	8192	33.7	0.961 [0.944, 0.976]	0.443	0.000

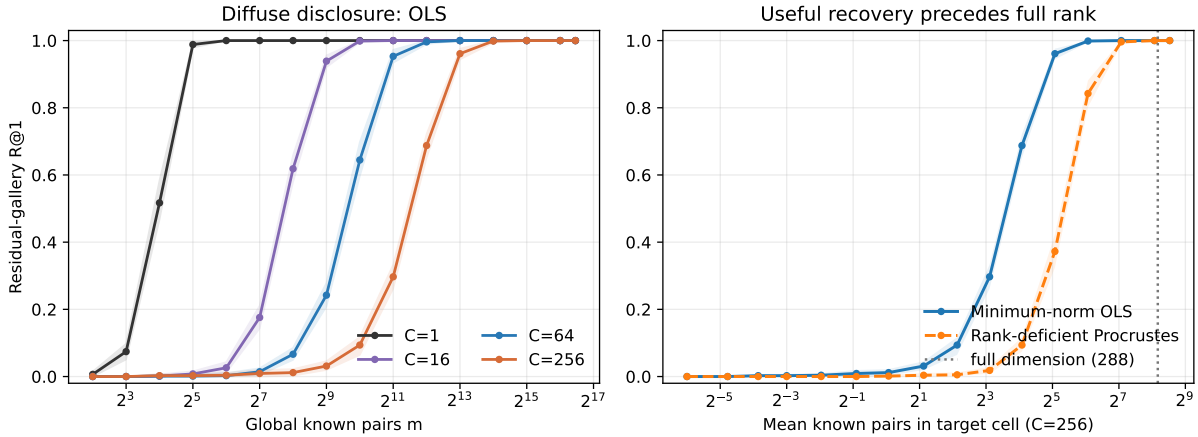


Figure 7: Partial residual alignment without a full-rank gate. Minimum-norm OLS extracts the observed anchor span and reaches high gallery R@1 far before the complete cell key is identifiable. Increasing C shifts the global disclosure curve by spreading anchors across cells; it does not create a hard threshold.

For a targeted observer, only the in-cell count matters: the OLS rows above indicate that approximately 32–36 well-spread pairs in the victim cell already give high residual-only R@1 in this gallery, independently of the global cell count. The experiment excludes the unchanged public prefix, norm and cross-release Gram channels; Section 7.6 shows that the full server view is easier to link.

Checkpoint-compatible text outcome under a strengthened observer. Embedding-space R@1 does not by itself say whether the recovered vector yields readable text. Experiment 29 therefore sends actual SHARD-derived views to the official public GTR-base VEC2TEXT inversion/correction checkpoints [3, 7], using their mask-aware mean-pooling path rather than the incorrect CLS extraction used by an earlier diagnostic. The corpus contains 2,400 cached AG News records and 600 controlled synthetic records, truncated to the checkpoint’s 32-token input. We fit a full 768-dimensional PCA and evaluate $(d_{\text{pub}}, C) = (192, 16)$ and $(384, 32)$, two independent cell keys per geometry, 12 targets from the globally largest public cell, eight correction steps, and nested $m \in \{8, 16, 32, 64\}$ in-cell pairs. The views are raw, prefix-only, a keyed residual treated as if its key were the identity, exact-key oracle, minimum-norm OLS, and rank-deficient Procrustes.

This is deliberately a strengthened, checkpoint-compatible stress test. The observer is granted the exact PCA basis V and mean μ , leaving only the cell key H_c unknown or partially learned; it is therefore stronger than the honest server in our primary threat model and isolates the outcome of residual key recovery. The official corrector expects native unnormalised mean-pooled GTR vectors (mean norm 1.047 here), so this case is not the literal unit-normalised e5 retrieval instance. Raw and exact-key oracle are identical geometry controls, not upper bounds on a non-monotone neural decoder. The intervals are conditional, descriptive bootstraps over two fixed geometry

designs and their targets, not population-level significance claims.

Table 13 gives the globally largest-cell outcome. Averaged equally over the two geometry designs, the unknown-key view lowers token-F1 from 0.665 for the exact-geometry control to 0.242. This is the positive no-anchor effect that survives the stronger raw control. It is not isolation: the public prefix alone retains F1 0.330 at $d_{\text{pub}} = 192$ and 0.536 at $d_{\text{pub}} = 384$. With only eight in-cell pairs, minimum-norm OLS reaches 0.343/0.557, essentially returning to the prefix-only outcome. Increasing the pair budget improves geometric cosine but does not make neural decoding monotone; at $m = 64$, F1 is 0.373/0.516. Rank-deficient Procrustes is weaker in this grid.

Table 13: Checkpoint-compatible GTR text outcome in the globally largest public cell (12 AG News targets per geometry, two keys). Each cell reports input cosine / token-F1 / BLEU. Values are descriptive means; raw and exact-key oracle are numerically identical, so they share one row.

Observer view	m	$d_{\text{pub}}=192, C=16$	$d_{\text{pub}}=384, C=32$
raw / exact key	—	1.000/.670/.207	1.000/.661/.214
prefix only	—	.888/.330/.042	.985/.536/.094
unknown key	—	.789/.152/.020	.971/.331/.047
minimum-norm OLS	8	.892/.343/.050	.986/.557/.114
minimum-norm OLS	64	.916/.373/.049	.990/.516/.107
Procrustes	64	.839/.192/.022	.979/.410/.062

A separately labelled, evaluator-selected cell makes the controlled PII diagnostic measurable; it is not an attacker-discoverable primary cohort. Across its duplicated key releases, exact repeated-name recall is .792 for raw, .458 for the unknown-key view and .708 for OLS at $m = 8$. These names are not unique (307 values among 600 records), and some target names also occur among anchors. More importantly, exact recovery of unique visible email addresses is 0/48 and of phone numbers 0/34 for *every* view, including raw and exact-key controls; card numbers have no visible denominator after truncation. The PII diagnostic therefore lacks a positive unique-ID control and supports no PII-protection claim. It is reported to prevent a favourable token-F1 result from being relabelled as sensitive-attribute security.

7.6 SHARD: the public channel leaks coarse structure

The baseline’s public PQ index sits on the full protected space and preserves 67% of exact top-10 neighbours (Section 7.1). SHARD’s public channel is only the short prefix u . We measure how much true (full-space) top-10 neighbour structure each public channel reveals (NN-overlap@10, e5-small, 100 000 documents, 1000 probes; Fig. 8). A $d/4$ prefix reveals NN-overlap 0.55 and a $d/8$ prefix only 0.36 (and 0.20 at $d/16$), against 0.76 for the baseline’s exposed SVD $k=d/2$ geometry. The public stage-1 channel thus discloses coarse/topic neighbours but not fine identity, which lives in the keyed residual. The pattern reproduces on e5-base: a $d/8$ prefix leaks NN-overlap 0.58 and a $d/4$ prefix 0.77, against 0.94 for the baseline geometry; the within-cell fraction is 0.55/0.46 at $C=64/256$.

We are explicit about the residual leak. Within a cell the keys cancel, so the server can compute exact same-cell similarities; the fraction of a document’s true top-10 neighbours that fall in its own cell—hence remain recoverable—is 0.50 at $C=64$ and 0.41 at $C=256$, against 1.00 for the baseline’s single global key (which cancels for *all* pairs). More cells localise the leak further, and the micro-key variant removes it entirely. SHARD therefore shrinks, but does not zero, the neighbour graph available to an honest-but-curious server.

Baseline: a PQ-only hardened store. A natural alternative to cell-keying is to publish product-quantization codes of the rotated store and rerank from them—a *PQ-only* defense. We

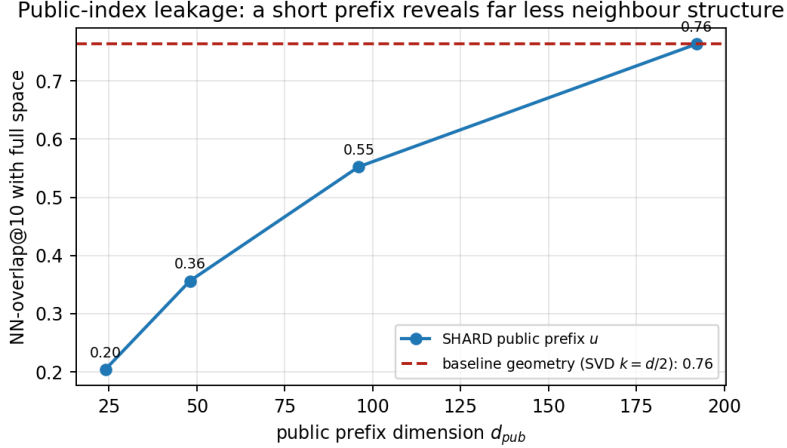


Figure 8: Public-index leakage (e5-small). The SHARD public prefix u reveals far less of the true top-10 neighbour structure than the baseline’s exposed SVD $k=d/2$ geometry; a $d/8$ prefix discloses about half as much.

Table 14: A PQ-only hardened store vs. SHARD (100 000 documents, 1000 probes). Rerank fidelity is the cosine of the channel used for scoring to the true vector (SHARD’s cell-keyed residual cancels exactly; PQ-only is lossy); NN-overlap@10 is the full-space top-10 structure recoverable from the public channel.

Encoder	Public channel	rerank fidelity	NN-overlap@10
e5-small	SHARD $d/4$ prefix	plaintext (1.000)	0.55
e5-small	PQ-only $M=24$ (24 B)	0.67	0.27
e5-small	PQ-only $M=48$ (48 B)	0.83	0.49
e5-base	SHARD $d/4$ prefix	plaintext (1.000)	0.77
e5-base	PQ-only $M=48$ (48 B)	0.67	0.41
e5-base	PQ-only $M=96$ (96 B)	0.83	0.63

compare it on the same data and leakage metric (Table 14); the result is an honest trade-off, not a clean win. At small code sizes PQ-only leaks *less* coarse neighbour structure than SHARD’s $d/4$ prefix (NN-overlap@10 0.27–0.63 vs. 0.55–0.77), because lossy quantization blurs the high-variance directions the prefix exposes. But that loss is the catch: PQ-only reranks from a reconstruction of cosine only 0.67–0.83 to the true vector, so retrieval quality degrades, whereas SHARD’s per-cell orthogonal keys cancel and rerank *exactly* (cosine 1.000)—the raw-nDCG recovery of Section 7.4 that PQ-only cannot match. The two are different points on the privacy/utility frontier: PQ-only buys a smaller footprint and lower coarse leak at a retrieval-quality cost; SHARD preserves the plaintext score and instead localises the residual leak to cells (and removes it at the micro-key limit, Section 7.6).

Micro-keys remove shared residual edges, not cross-release linkage. Per-document keys do eliminate shared-key residual comparisons: the fraction of a document’s true residual top-10 neighbours directly recoverable from same-key products falls from 0.25 at $C=64$ and 0.18 at $C=256$ to 0.00. That within-release result must not be confused with unlinkability.

Experiment 25 creates two independently re-keyed snapshots of the same e5-small records, independently permutes row handles, removes persistent document identifiers, and evaluates both a residual-only view and the full server view. At $N = 10,000$ the earlier raw cross-key cosine is indeed at chance, but stronger invariant features reverse the conclusion (Table 15). For $C=64$, monotone one-to-one assignment by residual norm links 99.6% of records and a permutation-invariant within-cell Gram signature links 99.9%. For micro-keys the Gram channel

Table 15: Cross-release linkage after independent row permutation (e5-small, clean snapshots, $N = 10,000$, mean over three seeds). The weak cross-key cosine control is near chance, but orthogonal invariants and the unchanged prefix link almost every record.

Keying	Linkage method	top-1	AUC
cell, $C=64$	cell-label chance	0.0064	0.500
cell, $C=64$	raw cross-key cosine	0.0057	0.501
cell, $C=64$	norm-rank assignment	0.9957	—
cell, $C=64$	Gram signature	0.9993	1.000
cell, $C=64$	unchanged prefix	0.9993	1.000
micro-key	global chance	0.0001	0.500
micro-key	raw cross-key cosine	0.0003	0.497
micro-key	norm-rank assignment	0.9943	—
micro-key	unchanged prefix	0.9993	1.000

Table 16: Common-document R@1 under insert/delete churn ($N = 10,000$, clean, three-seed mean). Values are the combined norm/Gram/neighbour signature; parentheses give Gram alone. Prefix and norm remain at $R@1 \geq 0.9996$ for all clean overlap conditions.

Encoder	keying	25%	50%	75%	90%	100%
e5-small	cell	.148 (.134)	.315 (.285)	.650 (.604)	.913 (.884)	1.000 (1.000)
e5-base	cell	.140 (.134)	.300 (.273)	.638 (.585)	.913 (.875)	1.000 (1.000)
e5-small	micro	.016 (.015)	.020 (.018)	.020 (.015)	.017 (.015)	.021 (.017)
e5-base	micro	.019 (.017)	.021 (.019)	.019 (.017)	.020 (.018)	.021 (.018)

vanishes, yet norm assignment still links 99.4%. The unchanged prefix links 99.9% in both cases. Cell-level Gram matching remains above 99.9% under fp16, int8 and a diagnostic 1% Gaussian perturbation. These are same-population snapshot results without insert/delete churn, not a universal linkage benchmark, but they decisively reject the former unlinkability claim.

Experiment 28 makes this game less artificial by giving each release $N \in \{2,000, 10,000\}$ rows but retaining only 25%, 50%, 75%, 90% or 100% of the underlying documents. Release-local row order is random, identifiers and overlap labels are hidden from the attacker, and the two releases contain disjoint insertion/deletion sets around the common subset. We repeat the game for e5-small and e5-base, three independent release pairs, clean float32, fp16 and release-local symmetric int8, using features standardised only from the two observed marginals. Thus no true pair or overlap label is used to build the linker.

Churn weakens Gram signatures because a row’s within-cell neighbourhood has changed, but it does not repair the unchanged channels. At $N = 10,000$ in clean releases, public-prefix R@1 for persistent rows is at least 0.9997 and residual-norm R@1 at least 0.9996 at every overlap. Even at 50% overlap, fp16 norm linkage reaches 0.9632/0.9782 for e5-small/e5-base. Int8 lowers those values to 0.4505/0.5522, but the corresponding pairwise AUC remains 0.9930/0.9951. Quantisation therefore makes exact scalar assignment less stable without erasing the signal.

An open-set, threshold-free mutual-nearest-neighbour rule applied to the public-prefix distance makes the churn effect visible in a different way. Its common-row recall stays essentially one, while precision is only 0.583/0.616 at 25% overlap and rises to about 0.996 at 90%. The low-overlap errors are false reciprocal matches among churn-only rows, not protection of persistent documents. Per-document micro-keys do destroy cross-release Gram invariance (roughly 0.016–0.021 R@1), yet they still leave the prefix and residual norm. The expanded game therefore narrows the earlier statement—Gram linkage depends on population overlap—while strengthening the central conclusion that SHARD releases are not unlinkable. Because clean and fp16 persistent prefixes also permit a near-zero distance threshold, this threshold-free MNN diagnostic is conservative for the unchanged-prefix channel.

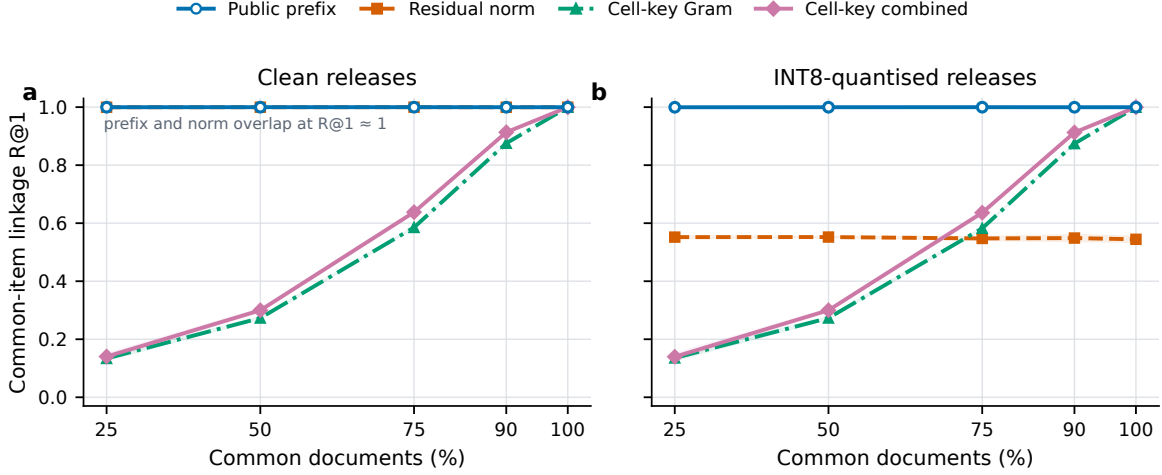


Figure 9: Linkage of persistent e5-base records across independently re-keyed $N = 10,000$ releases. Cell-key Gram structure weakens at low overlap and recovers as the common population grows; the stable prefix remains almost perfect. Int8 reduces exact norm R@1 but leaves a strong pairwise signal. Bands show one standard deviation over release pairs.

7.7 A formally calibrated Gaussian release

The earlier isotropic-noise sweep remains only a distortion diagnostic: it had no adjacency relation, clipping rule or privacy accountant. Experiment 27 replaces that misuse of terminology with an explicit one-shot mechanism. Databases have fixed size and are adjacent when one row is replaced; row participation and identifiers are public. Every embedding is clipped to $B = 1.000001$, a fixed bound from the encoder’s unit-normalisation contract rather than a corpus estimate. The concatenated release therefore has global L_2 sensitivity $\Delta_2 = 2B = 2.000002$. Independent Gaussian noise is added to every coordinate and each row is then unit-normalised as post-processing. For $\delta = 10^{-6}$, σ is obtained from the exact analytic Gaussian privacy-loss equation of Balle and Wang [23]; all five executable calibration checks recover the requested δ to numerical precision. Writing Φ for the standard-normal CDF, the solved equation is

$$\delta = \Phi\left(\frac{\Delta_2}{2\sigma} - \frac{\varepsilon\sigma}{\Delta_2}\right) - e^\varepsilon \Phi\left(-\frac{\Delta_2}{2\sigma} - \frac{\varepsilon\sigma}{\Delta_2}\right).$$

This gives central (ε, δ) -DP for document-vector content under the stated replacement adjacency when a trusted curator publishes the joint release. The same per-record calibration has a local interpretation only if each record owner clips and perturbs its own vector. It is not add/remove membership privacy, does not protect queries, and repeated independent releases compose. Centroids or PCA fitted on private release data would need their own mechanism and privacy accounting; here preprocessing parameters are public or independently trained.

We evaluate eight real BEIR/MIRACL encoder–dataset cases, 17 finite ε values, three noise seeds, every judged query, and 256 linkage targets per case against a clean native gallery of up to 5,000 rows after identifiers are removed. Confidence intervals use 2,000 hierarchical bootstrap repetitions. Table 17 summarises three operating points. At $\varepsilon = 1$, nDCG@10 is at most 0.011. By $\varepsilon = 512$, native-gallery linkage is already 0.962–0.995 while nDCG@10 is only 0.017–0.062. A strict utility match to corrected SHARD, defined by $|\Delta \text{nDCG@10}| \leq 0.01$ and $|\Delta \text{Recall@100}| \leq 0.02$, occurs in only three of eight cases on the finite grid, all at $\varepsilon = 32768$ and all with linkage R@1 at least 0.995.

The comparison is deliberately limited to retrieval quality. The Gaussian baseline scans the exact full corpus, whereas corrected SHARD uses two-stage $K = 200$ routing; latency and traffic are therefore not matched. More importantly, SHARD is not DP, and an empirical linkage rate is not a privacy budget. The useful conclusion is narrower: for this simple per-record Gaussian

Table 17: Analytic Gaussian release at $\delta = 10^{-6}$. Metric ranges are across eight evaluated cases and are rounded to three decimals; “matches” counts simultaneous matches to the corrected SHARD nDCG@10 and Recall@100 targets on the measured grid.

ε	σ	nDCG@10	Recall@100	link R@1	matches
1	8.4494	.000–.010	.000–.028	.000	0/8
512	.07240	.017–.062	.071–.244	.962–.995	0/8
32768	.007959	.297–.675	.250–.964	.995–1.000	3/8

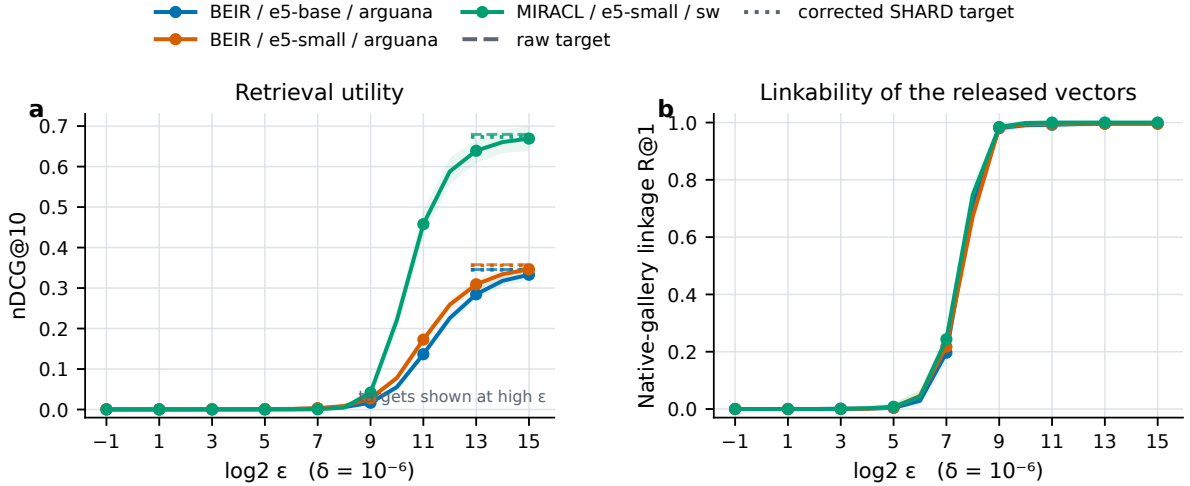


Figure 10: Utility and native-gallery linkability of the formal Gaussian release. Representative datasets are selected from non-private utility before examining any finite- ε result. Strong noise destroys retrieval; as utility returns, clean-reference linkage rises sharply.

release, every strict utility match occurs at $\varepsilon = 32768$ and has native-gallery linkage R@1 at least 0.995. It is a formal baseline with a different guarantee, not a proof that geometric keying dominates differential privacy.

A measured limitation: overlap reference lookup. The baseline allowed an overlapping reference corpus to recover exact source paragraphs after alignment (99.8% top-1, Section 7.1). The actual SHARD construction does not key the prefix: u_i is the stable stage-1 representation. Thus an attacker with an overlapping native reference does not need to recover a prefix key at all. The earlier optional keyed-prefix ablation reached 0.995 R@1 after 200 anchors on a 100 000-document e5-small reference; the unkeyed construction exposes at least as direct a channel, consistent with the 0.999 cross-release prefix linkage in Experiment 25. Residual cell keys and micro-keys do not close it. A coarse-cell-ID public channel or a client-private routing index could reduce this exposure, but their recall, storage and onboarding costs are not evaluated here.

8 Discussion

What compartmentalisation buys—and what it does not. Cell keys remove one globally shared residual orientation. Under diffuse disclosure, that shifts the required *global* pair budget because each cell sees only a fraction of the pairs. Experiment 24 shows the shift directly, but it also shows why it is not a privacy threshold: minimum-norm OLS uses the observed anchor span and re-identifies records from roughly 32–36 in-cell pairs, far below $d_{\text{priv}} = 288$. Micro-keys remove shared residual edges, but norms and the prefix remain stable. Thus C is a compartmentalisation–cost knob, not a security parameter. The positive result that survives is useful but narrow: corrected scoring preserves raw ranking, and diffuse evidence accumulates

more slowly per cell. The negative result is equally central: the stored view remains linkable and is not a cancellable template.

Encoder selection is part of the protocol. The baseline protective wrapper is metric-preserving only for the score it actually implements. Once query centring is corrected, the half-PCA gap is strongly encoder- and dataset-dependent: it is negligible for e5-base/NFCorpus but reaches 0.053 nDCG on MIRACL/Swahili with e5-small. Encoder and projection width should therefore be selected from corrected, graded relevance metrics on the deployment distribution; neither dimension nor σ_{rec} is a privacy or utility guarantee.

The role of the rotation R : obfuscation, not a primitive. We state this as plainly as possible: *the secret rotation is an empirical obfuscation layer, not a cryptographic primitive*. It contributes a defence channel orthogonal to SVD truncation, and even at $k/d = 1.0$ the unknown- R regime reduces off-the-shelf Vec2Text BLEU to the noise floor—but the paper also reports that a known- R attacker gains nothing beyond SVD and that a known-plaintext attacker recovers the orientation with standard Procrustes alignment. The aligned Vec2Text stress test does not recover text or typed PII, but its raw baseline is already near the reconstruction floor. Experiment 29 repairs that control in a separate, checkpoint-compatible GTR case: an unknown cell key lowers off-the-shelf reconstruction, but the public prefix and a small known-pair span restore part of it. Because that test grants V and μ , uses native unnormalised GTR vectors and does not train on SHARD outputs, a corpus-adapted or universal decoder is still untested. The rotation’s value proposition is therefore limited: it is a cheap obstacle for a weak unknown-orientation attacker, not a security guarantee. The scientific weight of the paper is the measured privacy/latency/accuracy trade-off and the documented failure modes, not the rotation.

What CKKS hides and what it does not. In the baseline prototype, CKKS hides the numerical scoring query and approximate score values from an honest-but-curious evaluator that lacks the secret key. Experiment 26 now executes that path for SHARD itself and shows that CKKS approximation preserves the measured order, but implementation evidence does not enlarge the security claim. CKKS does *not* hide active cells, ciphertext counts, the identifiers of candidates produced by stage-1 filtering, or the distances implicit in their order. Section 7.1 also shows that public PQ codes preserve substantial local geometry even before observing queries. An access-pattern attacker can therefore, over many queries, build a co-occurrence model of which documents are co-retrieved. Composing the pipeline with a PIR-style retrieval primitive (Tiptoe [40], SealPIR [41], OnionPIR [43], SimplePIR [42]) is the natural remedy; ours is engineered to be PIR-friendly because the ct-pt operations on the server are stateless and cluster-local.

Parameter selection. In the earlier baseline, the selected configuration is $\approx 1.7\times$ faster than the TenSEAL stock parameters at the same parameter-table security bound and within $\Delta\text{Acc}@1 \leq 1$ p.p. of baseline_proj. We reiterate that this speed-up is structural—ct-pt needs one multiplicative level, hence a shorter modulus chain—rather than a product of the learned surrogate, which only serves to extrapolate latency to unbenchmarked configurations or hardware. The selection runs offline at no per-query cost. These timings cannot be transferred directly to SHARD because its residual dimension, active-cell multiplicity and packing layout differ.

9 Limitations and Future Work

Intrinsic limitations. The most important limitation is now measured rather than hypothetical: cell-local orthogonal keys compartmentalise known pairs but do not create a de-anonymisation threshold. An observer needs full anchor rank to identify a generic complete key, yet a minimum-norm map reaches high residual-gallery R@1 from roughly 32–36 in-cell pairs. A targeted observer

can collect those pairs in one public cell, so increasing the global cell count does not improve this per-target case.

The stored representation is also linkable across key epochs. The public prefix is unchanged, every orthogonal transform preserves residual norm, and cell keys preserve the complete within-cell Gram matrix. Micro-keys remove the last shared-geometry channel but retain prefix and norm. Experiment 28 extends the same-population audit to two encoders, partial overlap and insert/delete churn. It still keeps the embedding of each persistent document fixed; text edits, stochastic re-embedding and encoder-version drift are not simulated. The numerical linkage rates should not be extrapolated to every deployment, but they are sufficient to reject unlinkability and cancellable-template claims for the stated construction.

The systems evidence is now concrete but still local. Experiment 26 measures SHARD transforms, CKKS encryption/decryption, ct-pt evaluation, serialization, point-sampled RSS, throughput, score error and ranking flips. It excludes preprocessing from online latency and does not include a real network, TLS, ANN service, concurrent clients or a packed multi-score response. TenSEAL executes this path on the CPU; the RTX 5060 does not accelerate CKKS. Most importantly, CKKS does not hide candidate identifiers, active cells, ciphertext counts or repeated-query access patterns.

The empirical scope remains bounded. The corrected utility audit covers two encoders and ten BEIR/MIRACL cells, with the largest graded corpus at about 297 000 passages. Partial alignment still uses e5-small residuals, while the churn audit covers e5-small and e5-base. The learned inversion experiment uses one GTR/Vec2Text checkpoint family, two fixed geometry designs and twelve targets per cell. To stay on the corrector’s input distribution it uses native unnormalised GTR vectors and deliberately grants the observer the exact PCA basis and mean; its bootstrap intervals are conditional descriptions, not population-wide significance claims. It cannot represent every generative or universal inverter. Membership inference and non-overlapping semantic reference lookup are not evaluated. The formal Gaussian release covers one simple mechanism under public participation and replacement adjacency; it is not an add/remove membership guarantee and does not privately learn PCA.

Priorities for future work. The first systems priority has shifted from implementing block-SIMD to fixing the response path it exposed. A useful client/server prototype should pack multiple candidate scores per response, include ANN routing and real RTT/TLS, and report throughput and tail latency under concurrent load. The present phase breakdown supplies a reproducible single-machine baseline for that work rather than a production-service forecast.

The first construction priority is to remove the invariants exposed by Experiments 25 and 28. One candidate is a well-conditioned asymmetric cell transform $z_i = A_c r_i$, $t_{q,c} = A_c^{-\top} r_q$, which preserves $t_{q,c}^\top z_i = r_q^\top r_i$ without necessarily preserving document–document norms or Gram geometry. This direction is related to asymmetric scalar-product-preserving encryption and must be positioned as a systems composition rather than a new cryptographic primitive. Any variant must be retested under the full server view, multiple release epochs, text/embedding drift, insert/delete churn and known-pair alignment.

Finally, the learned text experiment should be widened to corpus-adapted and universal inverters, more encoder families, entity/PII categories, membership inference and semantic lookup against non-overlapping corpora. On the formal privacy side, stronger baselines should include private preprocessing, add/remove membership adjacency and explicit multi-release composition rather than relabelling empirical non-linkage as DP. Both lines need a million-scale graded retrieval corpus before deployment-oriented claims are credible.

10 Conclusion

This study revises both the utility and the security interpretation of cell-keyed residual splitting. On utility, the decisive correction is simple: fit PCA on centred documents, retain centred coordinates for routing, but use the uncentred query for scoring. The resulting full score differs from the raw dot product only by a query-dependent constant. Ten BEIR/MIRACL cells confirm the identity and show that the earlier query-centring mismatch, not truncation alone, caused a substantial part of the reported baseline loss.

On security, cell keys do compartmentalise diffuse known-pair evidence, but the effect is gradual. Minimum-norm alignment recovers the anchor-span projection and exceeds 0.9 residual-gallery R@1 with about 32–36 in-cell pairs while the full orthogonal key is still underdetermined. The global disclosure budget grows as those pairs are spread over more cells, yet there is no hard d_{priv} de-anonymisation threshold and no corresponding secure choice of C .

The strengthened GTR outcome test gives the same conclusion in generated text. With the exact PCA frame disclosed but no cell-key anchors, mean off-the-shelf token-F1 falls from .665 to .242. The public prefix alone retains .433, however, and eight-pair OLS reaches .450. This is a bounded no-anchor benefit, not resistance to targeted or trained reconstruction.

Re-keying also fails the stronger unlinkability game. Residual norms, within-cell Gram signatures and the unchanged public prefix link almost every record across independently permuted releases. Insert/delete churn weakens Gram matching when overlap is low, but not the stable prefix or norm; micro-keys remove shared residual edges without closing those channels. The formal Gaussian baseline also shows why a different guarantee cannot be summarised by one empirical noise knob: useful retrieval returns only where clean-reference linkability is already high in the evaluated grid.

The implementation result is similarly specific. Real block-SIMD CKKS keeps the score order and sharply reduces query upload, but the current TenSEAL layout is slower because it emits one large encrypted response per candidate. SHARD should therefore be understood as a rank-preserving alignment-compartmentalisation mechanism with explicit leakage and a measured cryptographic query path, not as an unlinkable, cancellable, DP or cryptographically private document store. Breaking the observed invariants and packing the response path are the next construction and systems requirements before deployment-oriented privacy claims are warranted.

A Reproduced baseline analyses from the global-linear system

The security analyses of the global-linear baseline are the primary contribution of our prior work and are reported there in full [56, 57]; we do not reproduce them here and summarise only the conclusions the main text relies on. *Off-the-shelf VEC2TEXT* against SVD-truncated and rotated embeddings: an unknown rotation drops inversion BLEU to the noise floor, but *only* against a non-adaptive inverter—a known rotation, or the known-plaintext alignment of Section 7.1, removes the protection. *Lightweight projection baselines* (Gaussian and random-orthogonal): the choice of projection is an encoder-dependent utility trade-off, not a universal privacy primitive. *Matched-distortion Gaussian diagnostic (not a DP mechanism)*: matched-distortion noise can preserve more raw-neighbour structure than SVD at the same σ_{rec} , so σ_{rec} is not a privacy metric. *Aligned VEC2TEXT stress test* (Procrustes alignment then a generative corrector): no exact text or typed PII is recovered—but neither is it from the raw control, so this test bounds rather than certifies resistance. These are exactly the global-key weaknesses that motivate SHARD.

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